

- learning and memory problems: deficits in work, tasks, and commands, and
- getting stuck behind doors.

Behavioral signs are often the first, or only, signs of pain, illness, and of cognitive decline and so discerning the contributing factors can be a challenge for the family, and for the veterinary team. Senior pets may be less able to cope with stress, which may make them more susceptible to changes in their environment (for example a new cat, a house move). It is the job of the veterinarian and the rehabilitation team members to help to discern pain from stress and anxiety, even in a patient with CDS. When in doubt, treat for pain as a trial. Pain assessment, response to pain medications, and the overall wellbeing of the pet depend heavily on the measurement and assessment of the pet's behavior (Mathews, 2000). A wide range of behavior problems, ranging from avoidance, decreased activity, and inappetence to irritability, restlessness, and aggression could be due to underlying pain. In fact, any change from normal behavior and the development of new and abnormal behaviors can also be due to underlying pain or disease (Landsberg *et al.*, 2013b).

Monitoring both age-associated cognitive and physiological changes should be conducted at least annually in dogs (starting at 5–8 years for larger breeds and 8–10 years for smaller breeds) and cats (starting at 10–12 years) (Hammerle *et al.*, 2015). Clients with elderly animals will not always mention behavior changes during veterinary visits, so veterinary technicians should be asking clients about any changes in their pets' behavior whenever they see dogs from 8 years of age and cats from 10 years at the latest (ideally they should do this from puppy- or kitten-hood onwards!).

In cats, thorough medical and behavioral history is required for diagnosis. Cognitive and motor performance appears to decline from approximately 10–11 years of age for cats, but functional change in the neurons of the caudate nucleus in the brain have been observed by 6–7 years (Landsberg *et al.*, 2010).

Signs of Cognitive Dysfunction Confused with Weakness or Pain Issues

The following signs of cognitive dysfunction may be confused with the effects of weakness or pain:

- Reduced mobility—especially negotiating obstacles (thresholds, car entry)
- Urinary or fecal accidents in the house
- Change in appetite
- Decreased alertness
- Change in interaction with the family.

What can Influence Behavior in Geriatric Patients?

Behavior in geriatric patients may be influenced by many factors, including the following:

- *Anxiety*—CDS can increase anxiety and the likelihood of a patient developing fear- and anxiety-related behavior problems (Overall, 2013).
- *Reduced mobility*—If movement is difficult this can increase the likelihood of patient's toileting in inappropriate places because they are unable to reach a more appropriate toileting area in time. Animals with reduced mobility also find it harder to move away from people or other animals if they feel threatened, which will further increase the likelihood of them showing defensive aggression (Landsberg *et al.*, 2013b).
- *Restlessness*—CDS patients often find it difficult to rest, which can result in them being unsettled both during the day and at night (Landsberg *et al.*, 2013b). This can slow progress during rehabilitation.
- *Reduced interaction with owners*—Geriatric patients may not want to play or enjoy other interactions with their family, such as petting or going for walks (Overall, 2013). This can negatively affect the pet-client relationship, and if pain is unrecognized and/or untreated this may be a risk factor for relinquishment or euthanasia.
- *Polydipsia and polyuria*—Conditions associated with polydipsia and polyuria such as diabetes mellitus, hyperadrenocorticism,

Table 12.3 Feline mobility/cognitive dysfunction questionnaire.

My cat...	Yes	Maybe	No
Is less willing to jump down			
Will only jump up or down from lower heights			
Sometimes shows signs of being stiff			
Is less agile than previously			
Shows signs of lameness or limping			
Has difficulty getting in or out of the cat flap			
Has difficulty going up or down stairs			
Cries when they are picked up			
Has more accidents outside the litter tray			
Spends less time grooming			
Is more reluctant to interact with me			
Plays less with other animals or toys			
Sleeps more and/or is less active			
Cries loudly for no reason/to try to gain my attention			
Appears forgetful			

NB. Need to ensure there are no environmental reason(s) for these behavior changes.

Source: Adapted from Gunn-Moore (2011, 2014).

or chronic kidney failure will increase the likelihood of a geriatric patient house soiling or waking their owners at night to ask to be let outside (Landsberg *et al.*, 2013b; Overall, 2013).

- *Neurological and circulatory disorders*—Medical problems affecting the central nervous system (e.g., brain tumors) or the circulatory system (e.g., hypertension) can cause or contribute to cognitive decline (Gunn-Moore, 2011; Landsberg *et al.*, 2011).
- *Some medications* can also increase the likelihood of animals showing behavior problems. For example, corticosteroids can be associated with increased appetite, urine output, restlessness, and reactivity to stimuli, which can increase the likelihood of an animal showing problem behaviors including housesoiling, wandering, and pacing and also aggression to owners or other pets (Landsberg *et al.*, 2012).

Cognitive function screening questionnaires can be used to check for CDS. An example of a canine questionnaire can be found at: http://merrimackvet.evetsites.net/sites/site-3774/documents/CDS_checklist.pdf and an example of a feline mobility/cognitive dysfunction questionnaire is given in Table 12.3.

It can be difficult to differentiate between the signs caused by CDS and those caused by osteoarthritis-related mobility issues. Both conditions often occur concurrently in old cats and many of the treatments for one condition will also help the other.

Management Strategies to Improve Quality of Life

There are many management strategies that can improve the quality of life for elderly animals with mobility problems, including conditions associated with chronic pain,

sensory deficits, and cognitive dysfunction. Geriatric animals need to be able to navigate their surroundings easily. This can be particularly challenging for animals with medical problems affecting their mobility, including conditions associated with chronic pain such as degenerative joint disease and spinal problems. Conditions that reduce sensory abilities, for example vision and hearing loss or cognitive dysfunction, can be associated with impaired spatial awareness and navigational ability. Important resources that need to be accessible include food, water, comfortable resting places, toilet locations and, for cats, places to withdraw to or hide if they do not wish to interact with people or other animals in the home (Warnes, 2015b). Cats like to have food, water, and toilet areas kept separate. These must be easily accessible from the cat's resting area. If a cat spends time on different floors in the home it is sensible to locate a full set of resources, including a litter tray, on each floor.

Once resources have been located appropriately they should always be kept in the same places so animals can find them easily. If other animals are in the home, then it is imperative that there is no unnecessary competition for these resources (Table 12.4).

Other Considerations for Geriatric Pets

Introducing a new pet into the household can be extremely stressful for an older animal with mobility problems, and especially for one with cognitive dysfunction. Client may be better advised not to do this, especially with cats and any dog that does not have good social skills or is showing severe cognitive dysfunction.

Play can be associated with a positive emotional response and increasing aerobic activity will also boost circulation, increasing oxygen supply to the brain and muscles. Low-impact games such as gentle throw-and-fetch or search games are appropriate for most elderly dogs, and for dogs with vision loss. Search games to find food or toys

are particularly suitable. Short play sessions with fishing rod toys or toys that roll and/or make sounds will suit most cats. Older pets can become bored with toys quickly, so the toys need to be rotated every few days.

Dogs with mobility problems can be taken out in the car and then given a short walk in a new location, or can accompany owners on longer walks by riding in a modified baby stroller. Some elderly cats prefer to remain indoors, but if cats do want to go outside their safety can be improved by using a harness or possibly by fencing the garden to prevent the resident cat leaving and other cats entering. Screened back porches are excellent for environmental stimulation.

Animals with severe cognitive dysfunction or anxiety must have an environment that remains stable and is unchanging. Highly anxious animals, especially cats, may cope best when restricted to a single room containing food, water, a litter tray, and resting and hiding places. It is important to keep furniture and resources in the same places and to avoid big changes in the scent profile of the room, for example by not using strongly scented cleaning products, as these can be very challenging for cats. It also helps to maintain a consistent routine, ensuring that important events occur in the same order and at approximately the same times every day.

Common Presentation

Dr. Julia E. Tomlinson states, "There are two main aged populations in my practice. Those that are overweight who have varying mobility issues along with the potential for comorbidities; and the frail pet with weight loss, marked muscle atrophy, and weakness." Frailty syndrome is a decline in the body's functional reserve, lower energy metabolism, smaller muscle cells, and altered nervous, hormonal, and inflammatory functions (Figure 12.2). It is not a disease process but it does lead to increased susceptibility to disease and functional dependency. It is characterized by decreased muscle mass and loss of strength and endurance (Tomlinson, 2016).

Table 12.4 Suggestions for improving the environment and increasing access to resources for elderly cats and dogs.

Resource	Dogs	Cats
Food and water	Raising bowls off the ground will help dogs with joint and spinal problems to eat and drink more comfortably Non-slip matting underfoot will prevent dog slipping when eating or drinking	Need to be in separate locations Raise bowls off ground by a few inches to enable cats with joint and spinal problems to eat and drink more comfortably Cats used to being fed on raised surfaces, e.g., windowsills or worktops may need a ramp or steps to enable access, or food and water should be provided in more accessible locations
Toileting areas	Dogs with mobility problems may need to learn to use a toileting area closer to the house, or even be provided with a toileting area indoors, e.g., by placing puppy pads in a large tray Owners may need to encourage dogs to go to their toilet area regularly because they may no longer indicate when they need to toilet	Needs to be separate from feeding and drinking locations Cats with mobility problems will prefer large, low-sided litter trays, or equivalents such as gardeners' potting trays Finer grained litters are easier to stand on and dig than coarser litters Cats that have previously toileted outside may no longer be able to access these locations, and will need to be provided with litter trays indoors
Sleeping areas and beds	Beds should be comfortable and supportive, e.g., memory foam, easy for animal to get into and out of, and large enough for them to lie out flat if they wish Elderly animals can become cold very easily: sleeping areas should be kept warm especially at night in the winter. Electrically heated beds may be welcomed An Adaptil™ diffuser plugged in close to the bed may help reduce anxiety and help dogs settle better at night Items containing the owner's scent may also help some dogs settle better at night	Beds should be comfortable and padded, easy for animal to get into and out of, and large enough for them to lie out flat if they wish Cats prefer to rest in raised places, but animals with mobility problems may need ramps or steps in order to access these locations. Elderly animals can become cold very easily: sleeping areas should be kept warm especially at night in the winter. Electrically heated beds may be welcomed A Feliway™ diffuser plugged in close to the bed may reduce anxiety and help cats settle better at night
Moving around inside and outside home	Non-slip matting or carpet in locations of important resources and on the walkways between important areas can improve accessibility for elderly animals with mobility problems Non-slip ramps can help dogs navigate steep steps outside the home and also get into and out of cars Specially designed harnesses can be helpful for supporting dogs with mobility problems to enable exercise and access to toilet areas	Non-slip matting or carpet in locations of important resources and on the walkways between important areas can improve accessibility for elderly animals with mobility problems Most cats prefer raised resting places where they can feel safe and observe household activity from a safe distance. Providing ramps or steps may enable cats with mobility problems to continue to use withdrawal as a way of avoiding things that scare them Cats with mobility problems may no longer be able to use a cat flap so owners will need to let them in and out unless they prefer to stay indoors

Adapted from Warnes (2015b).

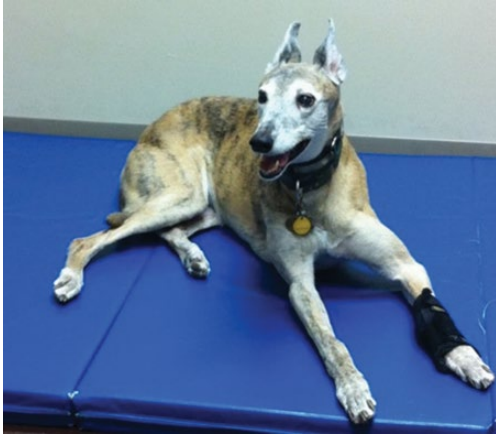


Figure 12.2 A frail canine patient. The patient is 15 years old. She has normal blood work and cognition but is weak and slow to move.

Geriatric patients often present to the rehabilitation veterinarian because of difficulties performing the activities of daily living, for example difficulty on stairs or getting into and out of the family vehicle. Another, often concurrent presentation is for management of chronic pain. The goal of the examination and treatment plan is to explore possible therapeutic options that will effectively improve strength, balance, and comfort without fatiguing the patient with resultant worsening home mobility problems. Slow, steady progress is the aim of any geriatric rehabilitation plan, and must include home assistance and nursing care regardless of the relative stability or functionality of the patient.

Health Benefits of Rehabilitation for the Geriatric Patient

The health benefits of rehabilitation for the geriatric patient should include (Edge-Hughes, 2009):

- improved daily function,
- reduced pain,
- improved strength,
- better oxygen transport and oxygen capacity,

- reduction in osteoporotic changes,
- reduction in blood pressure,
- improved breathing capacity,
- improved joint mobility,
- tranquilizing effect that reduces neuromuscular tension and anxiety, and
- greater capacity for human interaction.

Physical Rehabilitation and Function

Before any physical rehabilitation is undertaken, the patient must have an analgesic plan in place, most likely controlled with analgesic medication(s). There are four key areas of physical rehabilitation available to the geriatric patient: manual therapy; electrotherapy; hydrotherapy; and clinical/home exercise programs (Cotttriall, 2014).

Absolute precaution: “The patient should feel better, move better, and have better normal daily function when done with exercises. If they are more lame, sore after resting, or change transitions or posture for the worse, after exercises are done, then the plan needs to change” (McCauley, 2016a).

Manual Therapy the Veterinary Technician/ Nurse can Perform

Massage can provide relief from pain and spasm of muscles but should be exercised with caution in patients with low muscle mass. Elderly, frail patients may not enjoy massage or even light grooming. Active or passive muscle stretches may be prescribed and performed only in clinic for some patients, at least initially; other patients may be well enough for clients to be taught to use them at home. This can ensure flexibility and muscle length is maintained between treatments. Both joint range of motion and muscle length can also be supported with passive range of motion (PROM) exercises.

The PROM should be graded and only carried out after the client is cautioned to work within a limited range of motion if pain is likely to be an issue (Cottrill, 2014) (see Chapter 5 for more information).

Electrotherapy

Elderly patients may be slower to respond with discomfort during electrotherapy and so these modalities should be used with great caution. Thin skin, less muscle thickness and lower body fat can affect absorption of modalities such as laser and therapeutic ultrasound, so watch for discomfort. Reduced special senses and cognitive impairment may also mean that patient signals of discomfort will change or be more subtle (Figure 12.3). The therapist needs to be aware of this. These modalities are discussed in detail in Chapters 15–18.

- **Laser therapy:** This photobiostimulation increases cellular ATP and decreases nerve signaling of pain. Caution if the modality heats, patients with thinner skin and less muscle can experience burns more easily.
- **Obese patients** may “hold in” the heat of a laser in the fat layer and fat necrosis may occur.
- **Therapeutic ultrasound** uses thermal and non-thermal effects to reduce pain/spasm and increase tissue extensibility. Caution with patients that have low lean body mass; heating the periosteum with resultant damage and discomfort may occur.
- **TENS** (transcutaneous electrical nerve stimulation) is low-level electrical current which disrupts the normal pain perception pathways and can help to manage chronic pain. The first few sessions should always be supervised in clinic. Then a home unit can be prescribed with detailed instructions for use. If in doubt about possible discomfort during TENS, cease therapy and reassess with the veterinarian.
- **Shockwave therapy** uses high-energy sound waves to stimulate tissue repair and reduce neuropathic pain. This modality is very effective for arthritic pain but causes some discomfort during therapy sessions and temporary exacerbation of pain for a few days in some cases. Be careful about treating multiple joints in one session as mobility at



Figure 12.3 An elderly patient receiving neuromuscular electrical stimulation. The patient has thin skin, poor muscle thickness, and low body fat. The treatment must be applied with great care and the therapist should watch very closely for discomfort.

home may become temporarily much worse. Counsel clients about the risks of a pain flare and weigh the pros and cons with them.

- *PEMF* (pulsed electromagnetic field) therapy can help with pain, inflammation swelling, and wound healing. PEMF mats are available for home use (purchase or rental). Whole-body PEMF therapy should be avoided in patients with a history of seizures.
- *NMES* (neuromuscular electrical stimulation) is used to build muscle and has been shown to reduce muscle mass loss, increase its strength, and improve functional muscle use following orthopedic surgery (Cottrill, 2014). It may be painful over a severely atrophied muscle. Atrophy due to disuse maintains sensory innervation and so it is important to be cautious and err on the side of a lower level of muscle contraction in an elderly patient.

Approach to Exercise Therapy

All patients should benefit from warm-up exercises but they can be particularly helpful in stiff, elderly patients. The warm-up should aid muscle action and loosen stiff joints while avoiding too much active exercise and resultant fatigue before the targeted therapeutic exercises. The rehabilitation technician/nurse can give a brisk light massage up and down limbs and back to increase circulation and use range of motion to decrease stiffness. If the patient is very stiff, heat packs or warm towels/blankets should be placed on affected joints or whole body.

Once standing, cookie stretches and rhythmic stabilization exercises can be practiced. Assisted standing is used if the patient cannot stand on their own.

If the patient has low endurance, then give appropriate rest periods between repetitions and between exercises. Examples of exercise are walk to mailbox and back, rest 3 minutes

to an hour, repeat, rather than walk twice as far at one time. Sit to stand rest example: 1 sit to stand, walk 5–20 feet (1.5–6 m), 1 sit to stand, rather than 2–5 in a row. The patient may work up to 10–25 total.

Improving flexibility should be concentrated on early on in the program. Practicing range of motion and stretching is important. It is important to educate the client. Have owner keep a daily diary. Input information such as challenges – walk on uneven surfaces, over broom handles, core strength – sit pretty, diagonal leg lifts, flexibility – treat stretches, range of motion. Take special note of how they sit and get up from the down position.

Make sure the patient stays in the pain-free range.

Some geriatric patients may need assistance with mobility for therapeutic exercises. (Figure 12.4). Assistive devices include:

- Harnesses
- Lifts
- Slings
- Carts.

For more information about assistive devices see Chapter 10.

Therapeutic Exercises for Geriatric Patients

- *Weight-shifting/postural perturbations* help to improve strength, balance, and control, but the therapist should make sure that the patient is standing on a high traction surface. Be sure to counsel clients about gentle motions being effective and to avoid unbalancing their pet to the point of a fall risk. Gentle, more prolonged resistance to perturbation can be incorporated as the patient improves.
- *Cavalettis or poles* placed on the ground will help balance and coordination, even in sight-impaired patients. Go slowly and do the exercise in well-lit areas, avoiding bright sunlight as pupil constriction can

Figure 12.4 Some patients may need assistance with mobility. This patient has weak front limbs and the cart is used to aid mobility for longer walks which enhance quality of life.



impair vision further in patients with diminished sight.

- *Placing paws on targets* improves accuracy as well as subtle strengthening. Start with low targets such as a rubber mat and move on to a higher step, squishy foam pad, or even a balance disc.
- *Weaving* around obstacles in a figure-of-eight pattern can improve balance and control as well as spinal flexibility. Side steps are another strength and balance exercise. Competence with these two exercises can improve ability to turn in tight spaces in the house.
- *Backing up practice* (backwards walking) can also help home mobility.
- More capable pets can *balance* with one or even two legs lifted.
- Practicing sits ranges from assisted *transitions* (assisting joint motion and strength as needed) to holding a sit. Start with a small guided squat (half-sit) to standing exercise if needed. A raised platform can help a pet to keep limbs tucked in flexion and good posture for correct spinal alignment and muscle recruitment in a sit.
- *Active stretches for treats* (cookie reaches) will improve flexibility but also challenge balance and subtly strengthen. Stretches can be nose to hip, for example, down between front legs to arch back from a standing position (crunches), nose to rear toes.
- *Tail exercises* can help to improve rear awareness and strengthen the pelvic floor muscles. Stimulate tail motion with brushing the fur back at the tip of the tail, gentle squeezing, or even with happy talk.
- *Walking on uneven surfaces* can be a challenge for older pets but can also be used as a therapeutic exercise.
- Pets can also practice for *stair climbing by hill climbing* (serpentine up the hill first if easier) combined with step up and over low obstacles and progressing to practicing low stairs in the clinic or at home. Finally steeper steps are an option.
- *Getting into a vehicle* and onto furniture can be broken down into learned steps. A pet can be taught to step their front feet up on a raised level (equivalent of the bed, couch, or car) and then wait for assistance in the rear.

Special Considerations for Hydrotherapy

Hydrotherapy for the geriatric canine must be approached with caution due to the medical constraints that may prevail. It can be hugely beneficial as it will reduce concussive forces on joints and enable standing exercises to be carried out on patients with compromised mobility (Cottrill, 2014) (Box 12.8). The water treadmill ensures steady-paced

Box 12.8 Benefits related to hydrotherapy

- Decreases pain
- Osteoarthritis (chronic and acute on chronic)
- Cancer
- Decreases weight bearing stress
- Weight off the joints
- Muscle relaxation
- Enhances the ability of the muscles to contract and stretch
- Reduces edema in distal limbs
- Enhances coordination and balance
- Opportunity to strengthen core
- Area of weakness for a geriatric dog
- Increase metabolism
- Beneficial for overweight dogs
- Improves muscle tone and strength
- Works forelimbs, rear limbs, and trunk
- Metabolic weakness
- Water is a great medium for PROM/stretching
- Increased flexion of forelimb and rear limb compared to land treadmill or walking
- Wound healing if not full thickness
- Sanitizers clean “hot spots”
- Dry well after therapy and apply topical remedies or medications
- Increases superficial circulation

Source: Adapted from McCauley (2016b).

walking is achieved against resistance. Buoyancy aids, limb or body facilitation, and proprioceptive aids can all be applied to enhance the treatment. Only very short periods of therapy may be tolerated and respiratory rates must be monitored to avoid overexertion, especially in patients with compromised airway function such as those with laryngeal paresis. Muscle building and improvement in cardiovascular and aerobic fitness appear to be slower in elderly veterinary patients although gains can certainly be made. The goal is to maximize the therapy in terms of physiological and psychological outcomes. Watch for fatigue, both physical and mental.

Elderly patients are more likely to have some level of incontinence. Fecal contamination of water is possible, and urinary tract infections appear to be more common as full bladder emptying may not occur. It is a good approach to evacuate and express the bladder before therapy if there is potential for issues. Failure to achieve effective defecation if an elderly patient is too weak to posture can lead to anal gland issues. After hydrotherapy, many dogs need to urinate again. Be sure to allow them to walk outside to relieve themselves.

Increased abdominal pressure in patients with intra- or extra-abdominal masses can indirectly affect breathing. A full veterinary examination just prior to beginning therapy is recommended.

Swimming can be a good therapy for the elderly patient with the same caveats, but is probably more suitable for an elderly patient who has been a competent swimmer, as the stress and energy consumption levels will be too arduous for the inexperienced swimmer (Cottrill, 2014) (for more details see Chapter 19).

Physical Medicine and Rehabilitation for Patients in Palliative and Hospice Care

Veterinary patients in palliative and hospice care experience progressive medical diseases and can often benefit from physical medicine and rehabilitation (Tinkel and Lachmann, 2002; Downing, 2011). The patient with a progressive disease can often benefit from the application of physical medicine and rehabilitation techniques, not with the intention of curing the issue at hand, or necessarily reversing the disease process, but rather to maximize both comfort and function. Comfortable animals are more likely to continue to engage in normal, expected activities of daily living (ADLs). In addition, comfortable animals maintain their

relationships more easily with their human companions. (Downing, 2011). The most commonly applied physical medicine and rehabilitation techniques that lend themselves well to the hospice and palliative care setting include the following:

- Thermal modalities (cold/heat/therapeutic ultrasound)
- Massage
- Range of motion
- Stretching
- Chiropractic
- Joint mobilization
- Acupuncture
- Myofascial trigger point release
- Therapeutic laser
- Electrical stimulation
- Targeted pulsed electromagnetic field therapy (tPEMF)
- Therapeutic exercise.

For more details about each therapy see Chapters 5, 14–16, 19, 20, and 23.

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Conclusion

Rehabilitation objectives for the geriatric patient include managing pain, improving mobility and strength, providing appropriate assistive devices to promote independent ambulation, and modifying the patient's home environment to provide adequate traction, bedding, and obstacle-free space for ambulation (Starr, 2013).

Aids to Graceful Aging

There are five suggestions to helping the pet age gracefully. They are:

- manage weight,
- provide adequate nutrition – macro and micronutrients,
- recognize and treat pain,
- modify environment as needed to assist, and
- ensure appropriate exercise – strength, balance, flexibility.

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13

The Disabled Patient Part 4: Home Nursing Care

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Introduction

People face many challenges when caring for a disabled pet. Challenges can begin on the first day of the injury or disease, or they may develop over time despite the best efforts of both the veterinary team and owner to avoid or overcome them. Clients/caregivers need to gain a thorough understanding of their pet's diagnosis and the prognosis for recovery of function. Accurate prognosis can be very difficult to give in patients with neurologic issues. In our experience, recovery of an injured nervous system tends to be slow, with sporadic improvements and long plateaus and sometimes regressions (temporary or permanent). The patient that comes out of surgery for a disc fenestration/decompression able to walk when they could not do so before surgery

certainly has a good prognosis, however mild residual neurologic deficits can remain for life. Conversely, in a 2003 study by Olby *et al.* of dogs with spinal cord compression and loss of deep pain prior to surgery (traditionally thought of as having a poor prognosis; Lawson, 1971) a fair to good prognosis for return to ambulation was revealed.

Understanding that some pets are disabled for a brief recovery period, whereas others may remain disabled in the long term can help caregivers to be equipped mentally and physically at home and to think about home-life adjustments. An optimistic but pragmatic approach to the challenges ahead mentally equips people for the long-term care needs of their disabled pet. Understanding their pet's unique limitations and identifying how to help their pet function maximally with

adequate assistance can help to minimize the development of additional problems. For example, large dogs that are suddenly unable to support their own weight obviously can present a challenge when moving them around the home, even if the house is only one level. A home with stairs at each exit, along with up to the bedrooms and down to the basement will present significant obstacles. It may be extremely difficult to facilitate even the most basic of activities for the larger pet, whereas a small pet can be lifted and carried when needed.

The caregiver will need some help with caring for themselves. It is physically and emotionally draining to care for a disabled individual. Proper body mechanics should be explained in an effort to guard against injury (e.g., muscle strains).

Proper Body Mechanics

Body mechanics is a broad term used to denote an effort coordinated by the muscles, bones, and nervous system. It can either be

good or bad and can be directly related to the occurrence of back pains.

The following rules should be applied when transferring or moving your patients to protect your back:

- Keep the lower portion of your back in its normal position at all times.
- Move as close to the patient as you can.
- Do not twist your body. Always do a side step or a pivot.
- Set your feet into a comfortable and solid wide base of support when lifting.
- Keep your abdominal muscles contracted, bow slightly using the hips and squat.
- Keep the head upright and hold your shoulders up.

Pushing up from the knees and using your own momentum will help you lift the patient (Figure 13.1).

Elderly or disabled caregivers may encounter more challenges when caring for even the smallest of pets. Part of our job when providing rehabilitative care is to understand the logistics and limitations of the patient, of the home environment, and of the caregiver



Figure 13.1 Use of proper body mechanics.

themselves and then to apply this to a realistic home care plan.

Even under the best circumstances and with the most capable caregiver caring for the easiest patient, issues can develop that interfere with caregiver compliance. This, for example, may be a conflict of work schedules, lack of available helpers, or a case of compassion fatigue. It may be other outside influences. Sometimes, friends or acquaintances with strong but well-meaning opinions can have an effect which further burdens the caregiver with pressure to make a decision, or to do things differently. Caregivers need friends and family to provide mental and practical support in order to help them care for their disabled pet. When the support system of understanding family and friends is not available, caregivers can quickly become overwhelmed. They may even feel totally absorbed by their pet's problems, and as a result ultimately guilty for keeping their pet alive in their prolonged disabled state. Compassion fatigue is a natural response to a traumatizing effect experienced by someone close to you, the stress resulting from helping a suffering individual (Figley, 1995). The risk of compassion fatigue increases with ongoing exposure to suffering and with unexpected disruptions to the caregiver's life; this can be particularly common when there is a loss of hope for patient recovery (Sabo, 2011).

Preventing and treating compassion fatigue starts with identifying the risk and providing information to clients. A good place to start is the Compassion Fatigue Awareness Project (www.compassionfatigue.org) and Wrong Side of the Rainbow (www.wrongsideoftherainbow.com).

Respite care (see later in this chapter) is a way that the caregiver can take a short break from the constant daily care routine while knowing their pet is well cared for.

Teaching home care techniques can be difficult. Each person learns a little differently and it is important to approach the task with several different ways of teaching care (National Research Council, 2000). Verbal

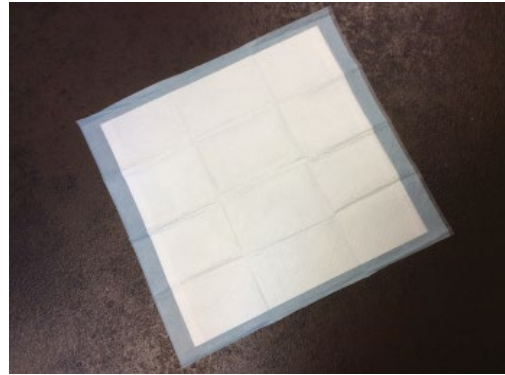


Figure 13.2 Absorbent diaper pads.

and written explanations are a good start, and handouts with pictures can be referred back to later. However, many people learn best by doing the task. Demonstrations should be followed by giving the caregiver an opportunity to go through the motions. This is essential to determine potential success with a procedure and to ease a nervous caregiver into an “I can do it” attitude. Analogies can be helpful. For example, the bladder can be compared to a water balloon when teaching someone how to express it. You can send home samples of materials needed, like absorbent “potty” pads (Figure 13.2) and this may help to get care started in the right way. A list of resources (websites) should be provided for mobility help, safety confinement, ramps, steps, or other care items.

Goals of Nursing Care

Maintaining Hygiene

This is a continuous process and with an incontinent patient can be viewed as a constant battle. Counsel caregivers that even a continent patient will have accidents, as a poorly mobile pet will not be able to easily signal their need to go outside. Gentle shampoo should be on hand for frequent bathing. Paper towels should be within reach of the caregiver but not the pet. Wet wipes can be useful but can be harsh and

Box 13.1 Hygiene care for the incontinent pet

Products list for maintaining hygiene:

- Gentle shampoo
- Astringent free wet wipes
- Absorbable diaper/pee pads (e.g., Chux)
- Emollient or barrier creams (diaper creams)
- Paper towels
- Waterproof bed liners

contain alcohol. Remember that frequent mechanical wiping can irritate skin, and sometimes rinsing is kinder and gentler. A hand-held shower attachment for the tub is a handy thing to have both inside (and outside when the weather is warm enough). Skin protectants and emollients can help to prevent issues and provide a barrier for areas of skin that are frequently wet or soiled. Stacking absorbent pads under the patient can allow for a quick easy clean-up in a time crunch situation; if an accident happens, simply remove the soiled top pad (Box 13.1). Multiple waterproof bed liners can help as they can be exchanged and protect underlying padding. Trimming excess hair from the area surrounding the anal opening helps to prevent stool blockage and ultimately constipation.

Assisting Mobility

The disabled pet will need varying levels of assistance in order to be mobile. Slings can be helpful and can be purchased in different sizes both with and without padding/linings for comfort. You can also make a sling by cutting the sides from a canvas bag. Simply cut one side of the bag, including the handle, from the top of the bag to the bottom and then the other side of the handle from top to bottom. Keeping the handles intact, you end up with a long strip of canvas with handles at each end (Figure 13.3). This simple homemade sling is lightweight, inexpensive, and easy to use. However, there is no padding so it should be used for brief

periods only. Harnesses that are padded are much more suitable for long-term use, though they should be removed daily and the skin checked for sores. An example is the Help 'em Up Harness™ (Figure 13.4).

It is important to counsel caregivers using any assistive lifting device about how to provide the minimum necessary assistance while allowing for some patient effort (Figure 13.5). Carrying the patient like a suitcase is uncomfortable for pet and caregiver and will not aid return of mobility.

Providing surface traction is a mobility aid that may be easily overlooked. Caregivers should be questioned about the home environment, including flooring. As a patient starts to regain some functional mobility, traction issues can prevent him or her from getting into a stand or from maintaining the limbs in adduction without slipping. Providing rugs on the floors, trimming nails and hair on feet (as better grip is obtained with foot pads), fitting boots or socks with grip bottoms (Figure 13.6), or toenail grips (Figure 13.7) can be instrumental in preventing unnecessary falls and allowing some mobility. Other mobility aids such as carts are discussed in Chapter 10.

Preventing Complications

Benjamin Franklin said “An ounce of prevention is worth a pound of cure” (Franklin, 1735). This statement can be applied to a disabled pet. The caregiver needs to be mindful of preventing contracture of muscles and connective tissue, including joints, protecting against skin sores and other injuries and treating pain that occurs from immobility and possible infection. Routine hygiene should always apply, for example removing feces in a timely manner and washing hands to avoid contamination spread.

When speaking of prevention of injury for the disabled pet, the whole home needs to be examined from the perspective of the patient. Confining the patient to a safe area away from potential falls or other injury is an important



Figure 13.3 Handmade sling.



Figure 13.4 Help 'em Up Harness.



Figure 13.5 Correct way to lift a dog in sling or harness.



Figure 13.6 Power Paws grip socks.



Figure 13.7 Dr. Buzby's ToeGrips™.

place to start. Remember that as a pet becomes more mobile, low barriers may be inadequate and further modifications may be needed. Bedding should be kept clean and dry and well padded. Clean dry bedding with waterproof pads beneath it is ideal, or absorbent disposable pads can be placed on top of the

bed. Placing mats in the rest of the patient area can help to prevent sores from a patient dragging themselves around. Yoga mats or padded play mats are ideal and are easy to keep in place so that they do not slide with the pet. They also wipe clean relatively easily.

A recent addition to the plethora of assistive devices for disabled pets is a “bag” which protects the rear limbs of patients who drag their rear around in a sit (scoot). The authors do not condone the use of this bag in most cases, because allowing a patient to gait pattern in a scoot reinforces this pattern as the “normal” gait and makes it harder to encourage walking with assistance (the ultimate goal of which is return to full mobility). However, in patients with no possibility of return to functional gaiting the bag can help to protect the skin of the rear limbs, but it must only be used when the patient is confined to a small space.

Recumbent patients should have their position changed, preferably every 4 hours. Prop aids, such as a pillow or rolled-up blanket or towel taped in a roll, can keep a patient in the sternal position to allow full inflation of both lungs and to improve venous return (Lamm *et al.*, 1994; Walther *et al.*, 1998). Upright positioning (sternal or assisted standing) for eating and drinking aids in digestion and can prevent problems with regurgitation and aspiration pneumonia. Food intake should be decreased to avoid weight gain due to less activity. A lower calorie food with the same micronutrient density and adequate protein intake to prevent muscle loss is ideal in our opinion. Lower fat, higher fiber diets can provide good satiety while keeping calories low and intestines healthy and regular. Conversely, sometimes we need to switch the patient to a low-residue food in order to decrease the number of times stool is passed or the amount of feces in each stool passed by an incontinent patient.

Preventing tissue contracture is achieved with manual therapies both in clinic and for the caregiver to perform at home. Therapies suitable for use at home include passive

range of motion (PROM), massage, and stretching. This touch is important as it helps pain management, strengthens the human–animal bond on both sides, and provides the motion that joints and muscles need to maintain health. Therapies such as these should be performed at home a minimum of twice daily as joint health is dependent on moving through a full range of motion every 8–12 hours (Matsuzaki *et al.*, 2013; Kojima *et al.*, 2014).

Maintaining Functional Elimination

It is our opinion that a regular schedule is the key to managing incontinence; using the natural gastro-colic reflex to stimulate stool passing can help to control accidents. We advise that after a patient eats, a caregiver should wait about 10 minutes then stimulate their pet by gently touching the perineal area with a cotton swab to stimulate anal sphincter contractions. This in turn will stimulate rectal contractions and passage of stools. If weather permits, it is good to take the patient outside to pass stools, even if they are minimally aware of their passing. Mimicking the old routine, with time of day and location can help to restore some conscious awareness of stool passing, if there is any residual sensation.

You and your supervising veterinarian can discuss changing dietary fiber in order to modify stools, either to more or less volume and firmer or softer consistency depending on the patient. This option should be tried before laxatives are introduced in the case of constipation, as an unfortunate side-effect of loose stools and urgency from treatment with laxatives can cause very difficult problems for the disabled pet.

Bladder emptying needs can vary. It is best to allow as much bladder function (contraction) as possible. For example, the caregiver can initiate urination with gentle pressure on the bladder to stimulate emptying and then allow the patient to continue urination until it slows. Once the stream of urine has slowed

and then stopped, the caregiver ensures complete emptying with full expression. Medications can help to improve bladder muscle and sphincter activity in some cases and will be prescribed by the veterinarian. Caregivers need to be counseled about the risk of urinary tract infections due to incomplete bladder emptying (Stiffler *et al.*, 2006). Having good access to drinking water so that urine production is adequate and assisting the flow if the stream is intermittent can help to prevent infections. Frequent urine samples should be taken. The provision of test dipsticks for home use is recommended as an unnoticed infection can cause further problems, including pain and more extensive infections.

Managing Pain and Comfort

Recognizing pain can be difficult for caregivers. Many signs of pain (such as being reluctant to move, stiffness getting up, and restlessness) are not apparent in an already immobile or less mobile pet. Helping caregivers to recognize subtle signs of pain can improve patient advocacy and so quality of life. A home pain scoring method is not always necessary but a checklist of some sort can help caregivers to maintain a pet's comfort. Keeping note of appetite, enthusiasm of greeting, sleep patterns, and demeanor in a daily diary is useful as it can detect patterns of comfort. Sometimes the most enthusiastic rehabilitation exercises aimed at regaining mobility can result in weakness and pain next day. The benefits of working hard to regain gait need to be balanced with everyday comfort and adequate pain management. Step one is enlisting the home caregiver to recognize and communicate that.

Pain scoring should be performed at every therapy visit. Do not assume that a patient with poor limb function has no pain just because they have reduced sensation (see Chapter 3 on pain management for more details).

A comfortable environment includes adequate access to food and water, clean padded surroundings, climate control, and good pain management. Bedding can be raised off

the ground (hammock bed) to allow ventilation (and drainage if urinary incontinence is present). A chilled gel liner placed under bedding can improve cooling, and a small fan placed just outside the confinement area can also provide some cooling. Be sure to avoid a constant direct breeze. Thin patients with low muscle mass from atrophy can benefit from heat-reflecting blankets.

Management of bed sores can be very challenging. Rotational use of multiple different bedding surfaces including memory foam, textured foam, and mesh hammock beds can help to change pressure distribution and prevent or relieve sores. Counsel caregivers about regularly inspecting the skin over bony prominences and alerting the rehabilitation team about any change, even hair loss. Early intervention can prevent full thickness wounds. Deep sores (full thickness skin defect) will need regular veterinary care. Foam pads can be applied or affixed directly to the skin around bony prominences. Bandages need to be changed daily, and education for home bandage changes will be of paramount importance. Digital pictures can be used to communicate progress (or lack of) when the patient is not in the clinic.

Assessing and Improving Quality of Life

It is an unfortunate reality that a disabled patient's life is somewhat restricted when compared to a patient with normal mobility. We have worked with disabled patients who become depressed. The good news is that the depression can be relieved with therapeutic intervention and some minor changes in home routine. Your supervising veterinarian may choose to provide antidepressant medications or to change pain medications that have a depressive effect. More importantly, home environment and mental stimulation can be changed (see Box 13.2).

Mental stimulation is key in preventing depression and boredom. Enriching the home environment is a relatively simple process. Mental stimulation can be improved by

Box 13.2 Questions to ask owners about the patient

Ask the caregiver(s) and their family questions about home behaviors so that you can use this to assess patient demeanor.

- Are they happy and interactive when they greet family members?
- Strangers?
- Does their pet seem to recognize and respond to loved ones, familiar voices and/or touch?
- Do they have normal sleeping habits?
- Are they attention seeking constantly due to boredom, or are they ignoring most people due to depression?

keeping the pet in the room which is the main hub of activity or by periodically changing rooms to change the scenery. Visual stimulation can be improved by propping the pet in front of a window. Smaller pets can be taken for walks in a stroller; larger pets in a wagon or cart. We have even had a disabled Great Dane patient who spent a lot of time outside in his large wagon cushioned with a dog bed. Toys can be stimulating for patients (who don't like to eat them). Storing toys and switching them from time to time can make old toys exciting again. Puzzle toys can be mentally stimulating and can be incorporated with meal time.

Motivation needs to be stimulated both at home and during in-clinic therapy sessions. Home rehabilitation exercises should include a special treat saved only for this time as this can improve motivation. Praise from family is also much needed for patient happiness and motivation. Purchasing a cart for the patient may improve the depressed patient's demeanor because some mobility is restored (see Chapter 10).

Improving mobility in the neurologic patient can take months or even longer. Neurologic patients usually have plateaus. Look for small triumphs to celebrate and praise. Small achievements are a big deal for

neurologically challenged pets and all of the rehabilitation team should show enthusiasm to help patient and caregiver motivation. Rehabilitation can have a great impact on the life of a disabled pet, especially with early intervention, even if return to full mobility is not achieved. The rehabilitation team will need to provide moral support for caregiver and pet. Creativity is key in both at-home and in-clinic therapies, but sometimes creativity is as simple as making the old new again.

Disabled patients that have orthopedic problems generally return to adequate function more quickly than neurologic patients. The therapy plan should include addressing pain from compensatory issues as well as the primary injury/surgery site. Home therapy can be a large part of this pain management using massage, PROM, and stretching; thermotherapy can also be provided at home. Caregivers need to be compliant with both restrictions in activity and with home exercise recommendations. Caregivers of pets with orthopedic problems may be less aware of, and therefore less careful with subtle disabilities. During therapy visits, the team can help caregivers to understand the need for continued restrictions in order to prevent further injury or worsening the existing injury. Counsel caregivers about the importance of following exercise recommendations in order to progress recovery in a timely manner and to achieve the best possible recovery.

Improving quality of life for the disabled geriatric pet can be challenging. Reduced vision or hearing may have already reduced environmental stimulation and interest; loss of sense of smell can affect appetite. Older patients can benefit from learning new behaviors and from changes in environment, even if this means a car ride rather than a walk in a new place. Patients may have been inactive before the onset of their disability and so can lack adequate strength and fitness needed to compensate for even minor disabilities. Muscle loss (sarcopenia) occurs with advancing age and with that strength is also lost (Bellows *et al.*, 2015a). Patients of very

advanced age can suffer from frailty syndrome—a decline in the body's natural reserves which results in increased susceptibility to disease and weakness (Bellows *et al.*, 2015b). Inactivity from injury or disability can lead to reduced strength and muscle mass, which feeds into poor mobility and can also lead to obesity. Even a small amount of weight gain can exacerbate lameness or weakness (Mlacnik *et al.*, 2006; Marshall *et al.*, 2010). Loss of functional reserve includes declining cardiovascular and respiratory fitness (Chen *et al.*, 2014). In our practice we see two main populations of aged patient: the thin, frail pet with poor muscle coverage and reduced appetite and the obese, unfit pet with or without concurrent disease. In both cases, the scales are tipped in the favor of morbidity and a small problem such as a low-grade bladder infection can be enough to result in a marked increase in weakness.

Caring for the aged pet at home involves supporting comfort and mobility. Comfort is addressed in many ways, from adequate footing and bedding, foot and nail care and managing weight, to providing pain relief. Moving water dishes or adding water sources near to where a patient is resting can help to maintain adequate water intake. Placing a mat in the kitchen where the pet eats can provide better traction and so eliminate some discomfort when eating. Raising the food bowl height can also improve comfort and ensure standing during eating (a good strengthening exercise). Mobility aids range from home environment modifications (ramps and steps to furniture, blocking off difficult stairs) to patient modifications (toe grips, harnesses).

Nursing care includes bowel and bladder care; keeping skin clean and unsoiled is important. Sometimes elderly pets need to eliminate more frequently and so require outside access more often. A geriatric pet may not be able to hold their tail up fully when eliminating, and may contaminate their skin. Posturing to eliminate may be difficult and so full evacuation may not occur, which predisposes to bladder infections and

constipation. Mobile disabled cats may have trouble getting into a normal litter box and so using a low-walled structure such as a baking pan can provide easier access. Anal sacs may become full and uncomfortable from chronic inadequate expression. Owners need to be counseled about subtle signs to look for, areas of skin to check, and tell-tale signs of problems such as urinary tract infections. Improving the quality of life in the geriatric pet is often about the little things. Just having a weak, old pet stand still for a minute can improve stamina, cardiac output, and respiratory function.

Modifying the Home Environment

As stated earlier, environmental enrichment is very important. Home modifications may or may not be necessary to aid mobility depending on pet size and the amount of assistance needed. Preventing slipping in low traction areas is necessary. A simple solution is rugs or runners placed on the floor surface. Ensure that the rug grips the floor and does not slide with the pet. Inexpensive mats and runners are available from home improvement stores. Confining the disabled patient is strongly advised when supervision is not available. A small room with high traction flooring and low to no furniture is ideal if the patient has some mobility. Yoga mats can provide temporary flooring with grip. A crate or kennel is advised for pets with inadequate or no motion in the rear limbs who pull themselves around dragging the rear (“scooting”). This confinement helps to avoid the scooting motion becoming the “normal” efficient gait for the pet, so reducing motivation to ambulate on four legs, even with assistance. Baby gates can be used to prevent access to stairs and to confine a pet to a small room. Higher barriers, such as a screen door, may be needed for larger and stronger pets. Ramps can be useful if a home has just a few stairs and also can be used for getting into the car. Be cognizant of the incline and

Box 13.3 Guide to caregiving at home

- Make a handout with pointers to help caregivers at home. Recruit previous clients for tips
- Include bowel and bladder care, bedding and hygiene advice
- Add helpful hints—“keep paper towels in every room for quick clean-up”
- State when a caregiver should contact the clinic for red flags such as bed sores

choose a suitable length of ramp. Carts are a great way of improving patient mobility, however owners need to be mindful of the width of the cart and wheels compared to the home hallways and potential limitations of actual space needed to maneuver a cart in the home. It may be better for carts to be used outside only.

Other owners of disabled pets are a great resource for home care tips. In our clinic we asked owners of long-term disabled pets to help us to prepare an “at home” guide to caregiving. This helped us to gain a fresh perspective and we prepared a handout for owners (Box 13.3).

Respite care is the provision of short-term accommodation in a facility outside the home in which a loved one may be placed. This provides temporary relief to those who are caring for family members, who might otherwise require permanent placement in a facility outside the home. Finding such a facility for a disabled pet can be a challenge. We have one boarding facility in our large metro area that provides the necessary level of care; the alternative is in a veterinary clinic. This highlights the need for more pet boarding facilities with nursing capacity.

Administering Medications to the Disabled Patient

Administering pills can be as simple as putting in them in the food bowl with daily rations, but sometimes giving medication

can be extremely challenging. Some pets can be very creative at avoiding actually swallowing their medicine. Custom treats designed to mask a pill can be useful. An alternative is to hide the pill in tasty people food. One thing many people do not know is that fat masks taste: a bitter pill may be easier to give in cheese or peanut butter (take care regarding patients sensitive to fats, such as those prone to pancreatitis). Liquid formulas can help. Medications can also be compounded into chewable taste tablets in some cases. Whole foods stores and coops often sell gel capsules that can be used to disguise a nasty-tasting pill. We often recommend giving two treats – the first treat is untainted by a pill and the second treat contains the medication. This way the pet may be less suspicious. It is best to follow a pill with water or food (especially one administered alone without a treat) in order to ensure that the pill does not remain in the esophagus, causing irritation. This is a particular issue with cats but may also be a problem in disabled pets who are immobile and unable to prop themselves up for long periods. A colleague administers pills to cats by following them with a “chaser” of water dripped into the mouth from a gauze sponge, which induces swallowing. If owners can restrain their cat then this is an excellent method of administration. With dogs, a snack is usually enough following pill administration.

Medication schedules can be challenging for many owners; your supervising veterinarian needs to take this into account. Some medications can be administered 2–3 times daily and maybe the middle day dose has to be skipped on some days due to work or life schedules. Choosing a pain medication takes into account many factors, one of which should be ease of administration at home, including caregiver schedule. This improves compliance. Making a medication chart that can be checked off or initialed when medications are given is a good idea in a situation where multiple caregivers are involved. Pill cases with daily cups that can be filled for a week at a time can also help. These pill cases

can be purchased for once, twice, or three times a day dosing. Both charts and weekly pill cases can help to prevent double dosing. Sometimes medication will need to be given in injectable form. Caregivers can be taught to administer subcutaneous injections at home, using sterile saline for practice during teaching sessions. A daily dosing chart is advised.

Side-effects vary with each medication and caregivers should be given detailed information any about potential side-effects related to the patient's medications. In the case of rehabilitation patients, the primary care veterinarian, specialist veterinarian, and supervising rehabilitation veterinarian may have all prescribed medications. The patient record needs to be kept up to date with all medications and any medication changes or dosage changes. This way any veterinarian involved in the case will be aware of potential side-effects and interactions. As the supervising rehabilitation veterinarian will see the patient on the most frequent basis during in-clinic rehabilitation, this is the person best suited to manage and oversee all medications, and to make changes regarding pain management or other concerns. The rehabilitation veterinarian will always keep the primary care veterinarian informed of any changes.

As a side note, it is best to avoid changing medications just before a weekend, when the clinic staff are not easily available to answer questions about medications and potential side-effects. The emergency veterinarian on duty may not be familiar with the case.

In-Home Hospice Care

Hospice care at home is nursing care for a patient in the final stages of life or a disease process that will end in loss of life. Caregivers of disabled pets will be already equipped to provide for their pet's needs when the end of life is near as they have been previously providing nursing care, often including bowel and bladder care. Practical and

Box 13.4 Compassion fatigue

Compassion fatigue is fatigue, emotional distress, or apathy resulting from the constant demands of caring for others. For more information and resources visit www.compassionfatigue.org

emotional support through home visits from veterinarians is variably available. The International Association for Animal Hospice and Palliative Care can provide more details (www.iaahpc.org) (Box 13.4).

Remember that even if it is no longer feasible to transport the pet for regular in-clinic care, recheck visits every 2–3 weeks can be a big help to aid home care. Providing

support via telephone or email can bridge some of the gap between visits (though no veterinarian can make a full diagnosis without the patient present, ongoing care can be provided remotely to some extent).

Conclusions

Nursing care is a large portion of the rehabilitation veterinary technician's job. As an animal's nurse you are their advocate and care provider both in clinic and at home through your education of the primary caregiver and their family. A successful rehabilitation plan should always include consideration of the home environment and necessary adaptations and nursing care.

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Resources

- Back on Track heat reflective mesh sheet or blanket (www.backontrackproducts.com)
- Chillow. Soothsoft Innovations (www.chillow.com)
- Discount ramps (www.discountramps.com)
- Pill Pockets (www.greenies.com)
- Pill Pals (<https://www.henryscheinvet.com>)

14

Modalities Part 1: Thermotherapy

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Introduction

Thermotherapy consists of application of heat or cold (cryotherapy) for the purpose of changing the cutaneous, intra-articular, and core temperature of soft tissue with the intention of improving the symptoms of certain conditions. Thermotherapy is a useful adjunct for the treatment of musculoskeletal injuries and soft tissue injuries. Using ice or heat as a therapeutic intervention decreases joint and muscle pain. Heat and cold have opposite effects on tissue metabolism, blood flow, inflammation, edema, and connective tissue extensibility. Thermotherapy can be used in rehabilitation facilities or at home (Brosseau

et al., 2003; Nadler *et al.*, 2004; Hurley and Bearne, 2008; Petrofsky *et al.*, 2013a).

Purpose

The goal of thermotherapy is to alter tissue temperature in a targeted region over time for the purpose of inducing a desired biological response. The majority of thermotherapies are designed to deliver the thermal therapy to a target tissue with minimal impact on intervening or surrounding tissues. The use of superficial thermal agents aims to decrease pain, alter blood supply (e.g., reduce swelling), and temporarily

change tissue properties (e.g., extensibility). Extensibility means that muscles can be stretched to their normal resting length and beyond to a limited degree. They are often the most convenient modalities as they are readily available, involve minimal expense, and are frequently safe to use as part of a home treatment program (Niebaum, 2013). The primary goal of any thermal modality is to facilitate the rehabilitation plan of regaining maximal function (Dragone *et al.*, 2014).

Heat

By increasing the temperature of the skin/soft tissue, the blood flow increases by vasodilatation. Heat increases oxygen uptake and accelerates tissue healing. It also increases the activity of destructive enzymes, such as collagenase, and increases the catabolic rate (Kellogg, 2006) (Boxes 14.1 and 14.2).

Cold

By decreasing the temperature of the skin/soft tissue, the blood flow decreases via vasoconstriction. It will be followed afterwards by a vasodilatation. Tissue metabolism temporarily decreases under the influence of cooling affecting neuronal excitability, inflammation, and conduction rate (Boxes 14.3 and 14.4). Tissue extensibility also decreases under the influence of cooling due to increased tissue viscosity and decreased mobility of non-elastic tissues (Petrofsky *et al.*, 2013b). At joint temperatures of 30°C (86°F) or lower, the activity of cartilage-degrading enzymes, including collagenase, elastase, hyaluronidase,

Box 14.1 Effects of local application of heat – decreases

Local application of heat decreases:

- Blood pressure (if heat is applied for a prolonged time or over a large surface area)
- Muscle spasm
- Pain

Source: Adapted from Hayes (1993).

Box 14.2 Effects of local application of heat – increases

Local application of heat increases:

- Body temperature, respiratory rate, and heart rate if heat is applied for a prolonged time
- Capillary pressure and permeability (which can promote edema)
- Leukocyte migration into the heated area
- Local circulation (promoting healing in subacute and chronic inflammation)
- Local metabolism
- Muscle relaxation
- Tissue elasticity

Source: Adapted from Hayes (1993).

Box 14.3 Effects of local application of cold – decreases

Local application of cold decreases:

- Blood flow because of vasoconstriction
- Edema formation
- Hemorrhage
- Histamine release
- Local metabolism
- Muscle spindle activity
- Nerve conduction velocity
- Pain
- Spasticity
- Response to acute inflammation or injury

Source: Steiss and Levine (2005). Reproduced with permission of Elsevier.

Box 14.4 Effects of local application of cold – increases

Local application of cold increases:

- Connective tissue stiffness (with decreased tensile strength)
- Temporary muscle viscosity (with decreased ability to perform rapid movements)
- Activation threshold of tissue nociceptors

Source: Millis (2015). Reproduced with permission of Elsevier.

Table 14.1 Pathophysiologic effects of topical modalities.

	Heat	Cold
Pain	↓	↓
Spasm	↓	↓
Metabolism	↑	↓
Blood flow	↑	↓
Inflammation	↑	↓
Edema	↑	↓
Extensibility	↑	↓

↓ = decrease; ↑ = increase.
Source: Adapted from Nadler *et al.* (2004).

and protease, is inhibited. The decreased metabolic rate limits further injury and aids the tissue in surviving the cellular hypoxia that occurs after injury (Bleakley *et al.*, 2004; Hubbard and Denegar, 2004) (Table 14.1).

Types of Thermotherapy

Heat

Heating agents are classified as superficial or deep heating. Superficial heating agents penetrate up to approximately 2 cm depth, whereas deep heating agents elevate tissue temperatures at depths of 3 cm or more (Steiss and Levine, 2005). Heat sources are classified as radiant, conductive, or convective. An infrared lamp is an example of a radiant superficial heating device, a hot pack is an example of a conductive superficial heating device, and a whirlpool is an example of moist heat delivered by conduction and convection.

Superficial Heat

Superficial heating agents may include: hot packs, heat wraps, hosing with warm water, whirlpools, paraffin baths, circulating warm water blankets, electric heating pads, and infrared lamps (Dragone *et al.*, 2014). Another common form of superficial thermotherapy is heated beds for dogs, which

Box 14.5 Indications for heat therapy

- Subacute and chronic traumatic and inflammatory conditions
- Decreased range of motion attributable to stiffness and/or contracture (basis for the principle of “heat and stretch”)
- Pain relief, because heat may render sensory nerve endings less excitable

Source: Steiss and Levine (2005). Reproduced with permission of Elsevier.

have been used for comfort or conditions such as arthritis (Box 14.5).

Deep Heating

Deep heating agents include therapeutic ultrasound and shortwave diathermy. Diathermy is a therapeutic treatment most commonly prescribed for joint conditions such as rheumatoid arthritis and osteoarthritis. In diathermy, a high-frequency electric current is delivered via shortwave, microwave, or ultrasound to generate deep heat in body tissues (Giorgi and Crucik, 2013). It is unknown to this author if diathermy is used in veterinary medicine. The patient would need to be very still because movement can change amount of heating. Therapeutic ultrasound is dealt with in detail in Chapter 16.

Superficial Heating Agents

Hot Packs Hot packs come in a variety of sizes and shapes. They can be heated sacks with a canvas covering that may be filled with cracked corn, beans, bentonite (hydrophilic silicate gel), or other inert materials. Other hot packs are electric heating pads, damp microwaved towels, or circulating warm water blankets (Box 14.5).

Water Water may be applied directly to the area being treated. Towels heated with warm water may be used, or the affected segment may be immersed in a warm water bath or a whirlpool. If a whirlpool is chosen, be mindful that systemic heating with a warm bath may decrease blood pressure and increase the

heart rate (Dragone *et al.*, 2014). Whirlpools have the advantage of also providing increased hydrostatic pressure to submerged body parts, which may reduce edema. Increased hydrostatic pressure helps to increase lymphatic and venous flow from a distal to proximal direction. The temperature of a whirlpool is based on the needs of the individual animal. For example, patients with chronic conditions may be treated with warmer water than patients with more acute disorders.

A system employed by Dr. Rick Wall at The Center for Veterinary Pain Management and Rehabilitation (Woodlands, TX, USA) is bio-electric whirlpool therapy. This is a treatment combining the benefits of hydrotherapy using VetSystem's™ warm water whirlpool and the pain management benefits of electrical stimulation using ASP's Omnistim 500 Pro®. When the two are combined it gives a time-saving full-body treatment. This offers an avenue in providing pain relief and shortened recovery times for a variety of conditions. In their clinic, with a combination of physical medicine, the bioelectric whirlpool has assisted in the treatment of animals recovering from orthopedic joint surgeries, osteoarthritis, edema, chronic pain, and acute injuries such as sprains, tears, and muscle fatigue. The warm water and jets from the whirlpool deliver a comfortable

massage to the animal, increasing blood flow, promoting endorphin release and general relaxation (Figures 14.1–14.3).

Paraffin Baths This procedure is uncommon in veterinary medicine because it is messy. Details can be found in Rothstein *et al.* (2005).

Infrared Lamps Infrared lamps emit infrared light that penetrates the skin to promote increased blood flow and circulation. This method of thermotherapy is used to warm large areas of the body. The lamps are positioned 30–40 cm from the affected area. The therapist should place his or her hand under the heat lamp for several minutes at the desired height to be sure that the temperature is comfortable. Because the patient needs to remain stationary during treatment, this method of heating is uncommon (Dragone *et al.*, 2014).

Ceramic Agents Some products use ceramic to reflect the wearer's body heat and so warm a body part. Back on Track® products are used to keep muscles warm in canine or equine athletes, but are also used for the symptomatic relief of painful stiff joints. These products are made from a fabric which contains polyester thread embedded with a fine ceramic powder. They have been used in the authors' clinic to keep the muscles of

Figure 14.1 Whirlpool bath control.
Source: Courtesy of Kari Koudelka.





Figure 14.2 Vet Systems whirlpool bath and control. *Source:* Courtesy of Kari Koudelka.



Figure 14.3 Kari Koudelka with a patient in the Bioelectric Whirlpool.

athletic dogs warm between activities and some clients have reported pain relief for their older dogs when wearing the products. The company claim that the reflection of body heat is through far-infrared waves.

It has been claimed that far-infrared waves (Masuda *et al.*, 2005; Yu *et al.*, 2006):

- increase blood circulation and oxygen supply to tissues,
- reduce inflammation and swelling in muscles and joints,
- ease muscle pain and tension,
- speed up the healing process, and

- warm the muscles prior to exercise, thereby reducing the risk of strain and injury.

Cryotherapy

Cryotherapy is the therapeutic application of cold in rehabilitation and physical therapy. Cold can be applied through a variety of mechanisms including cold packs, ice massage, cold water baths, mechanical and electrical compression units, and vapocoolant sprays. Cryotherapy can be used throughout the rehabilitative process to

Box 14.6 Indications for cold therapy

- In animals with acute injury, rest and ice can be readily used. Application of cold to minimize postsurgical swelling is also recommended
- The reduction in pain and inflammation with cryotherapy may lead to increased range of motion in affected joints
- Local cold application may reduce spasticity in spinal cord disorders
- Cooling may inhibit the extension of tissue damage with thermal burns, with greatest benefit occurring when a coolant is applied immediately after burn injury

Source: Hanks *et al.* (2015). Reproduced with permission of Elsevier.

mitigate negative effects of inflammatory responses (Hanks *et al.*, 2015). The sensations reported by people after ice application are an initial sensation of cold followed by burning, aching, and eventual numbness (Rintamäki, 2007). Cold penetrates deeper and lasts longer than heat because of the decreased circulation resulting from cold application (Steiss and Levine, 2005). The primary method of providing physical control of pain and inflammation in the immediate postoperative period is cryotherapy (e.g., application of ice). Cryotherapy is used during the acute phase of tissue injury and healing to mitigate the effects of tissue injury. Cryotherapy is also used after exercise during rehabilitation to minimize adverse secondary inflammatory responses (Millis, 2004) (Box 14.6).

Cryotherapy may be achieved by the following physical mechanisms (Dragone *et al.*, 2014):

- Conduction (ice/cold packs, iced towels, ice massage, contrast bath, cold compression units)
- Convection (cold baths)
- Evaporation (vapocoolant sprays).

Cryotherapy Agents

Ice or Gel Packs Examples of ice packs/cold packs are crushed ice placed in a moist towel; a plastic bag wrapped in a moist towel; or water and rubbing alcohol of a 3:1 ratio in a sealed plastic bag and placed in a freezer (this should result in a slush-type consistency) (Burnett and Wardlaw, 2012). When using a plastic bag filled with ice or a gel pack, cover it before application with a thin, wet layer of fabric to improve temperature exchange between the tissue and cold agent. Do not use a thick layer of fabric, such as a towel or blanket, because therapeutic temperatures may not be reached in deep tissues.

Ice Massage Freezing water in cups or any other form of cylinder can be used to form an ice massage medium. To apply the massage, the therapist holds onto the cup, exposes the ice surface, and puts it in direct contact with the patient's skin. Treatment time is generally 5–10 minutes or until the affected area is erythematous, slightly pink or red, and numb (assess by pricking with a small gauge needle). This technique can be useful for small, irregular areas (Dragone *et al.*, 2014). Ice popsicles can also be made by placing a handle, such as a tongue depressor, into the water before freezing. The general recommendation for treatment duration is 5–10 minutes (Niebaum, 2013).

Towels in Ice Water Multiple towels must be used because a single towel will not maintain adequate therapeutic temperatures long enough to be effective. An effective method of using towels is to immerse two towels in ice water and alternate application when the towel being used becomes too warm for therapeutic effectiveness (Bockstahler *et al.*, 2004).

Contrast Baths This technique is used by alternating cold and warmer water. It is generally used to decrease edema because of the alternating vasoconstriction and vasodilation. An example of this would be immersion in the cold water for 3 minutes

and then immersion in warm water for 1 minute (Dragone *et al.*, 2014). The treatment therapy should end with cold.

Cold Compression Units These devices are commercially available and use a combination of controlled pressure with a continuous flow of cold water to help minimize swelling and pain postoperatively. The device usually consists of a sleeve with tubing running throughout that alternately circulates cold water and air. The combination of compression and cooling is effective in treating tissues in the acute phase of healing. An example of such a unit is the Game Ready®. This technology was developed for canine and equine patients from a human technology. Detailed information can be found at www.gamereadyveterinary.com/ (Figure 14.4).

Cold Bath/Immersion In this type of application, the patient typically stands with the affected limb immersed in a container of cold water at 2–16°C (35–60°F), and the treatment generally lasts for 10–20 minutes. Rapid and significant tissue cooling occurs, but is difficult to apply because of poor patient compliance (Hanks *et al.*, 2015). This therapy is used more frequently with horses, especially for soaking feet that are possibly laminitic. Andrew Van Eps and Christopher Pollitt studied the best methods for

adequately cooling the feet of at-risk horses (Van Eps and Pollitt, 2004). It is generally accepted that cold therapy should be continuous for 48–72 hours during the entire developmental phase and for another 24 hours beyond the end of clinical signs of the primary disease. It is important to include the foot, pastern, fetlock, and distal cannon region in the cold therapy for the best results and to maintain intimate cold contact with the limb to achieve therapeutic temperature ranges (Figure 14.5).

The suggested range for therapy is from 5 to 10°C (41–50°F). To achieve intimate contact, crushed ice is preferable to ice cubes. Ice wraps were not suitable because of their size and the air space created between the wrap and the leg. Dry cold therapy provided by a Game Ready® Equine system (CoolSystems Inc.) provided the most intimate contact and also provided intermittent pressure that may aid in blood and lymphatic flow. Treatment should continue 24 hours beyond resolution of the primary disease. New studies are demonstrating the benefit of cryotherapy in the early acute phase (Hyman, 2015).

Vapocoolant Sprays These sprays are used for a brief and rapid effect when applied to the skin. An example is ethyl chloride spray. This can be a difficult modality to apply to dogs because of the hair covering



Figure 14.4 Game Ready System cryotherapy and compression unit.



Figure 14.5 Boot for cryotherapy in a horse. Source: Courtesy of Dr. Chris Pollitt.

the skin surface. Also, some of the commercially available sprays may be harmful to animals if they contact an animal's eyes or are ingested. They are used in humans for treating myofascial trigger points (Dragone *et al.*, 2014). They are excellent for venipunctures once the skin has been shaved.

Mechanism of Action

Skin blood flow is controlled by two branches of the sympathetic nervous system: a noradrenergic vasoconstrictor system and a cholinergic active vasodilator system (Charkoudian, 2003). These dual sympathetic neural control mechanisms affect the major aspects of thermoregulatory responses over most of the body's surface. During periods of hypothermia, falling core and skin temperatures

lead to reflexive increases in sympathetic active vasoconstrictor nerve activity to reduce skin blood flow and conserve body heat (Sluka *et al.*, 1999). During periods of heat stress, increasing core and skin temperatures lead to reflexive increases in sympathetic active vasodilator nerve activity to increase skin blood flow (Charkoudian, 2003). Heat has an analgesic effect via heat-sensitive calcium channels, which respond to heat by increasing intracellular calcium. This generates action potentials that in turn increase stimulation of sensory nerves, causing the feeling of heat in the brain. These channels are part of a family of receptors called TRPV receptors. TRPV1 and TRPV2 channels are sensitive to noxious heat, while TRPV4 channels are sensitive to normal physiological heat (Holowatz *et al.*, 2005). Their multiple binding sites allow a number of factors to activate these channels. Once activated, they can also inhibit the activity of purine pain receptors (Liu and Salter, 2005). These receptors, called P2X2 and P2Y2 receptors, are mediated pain receptors and are located in the peripheral small nerve endings. For example, with peripheral pain, heat can directly inhibit pain. However, when pain is originating from deep tissue, heat stimulates peripheral pain receptors which can alter what has been termed "gating" in the spinal cord and reduce deep pain.

Studies have suggested that temperature can affect the exchange between Ca^{2+} and Na^{+} in neural cells (Swenson *et al.*, 1996). An increase in both pain threshold and pain tolerance with the use of cooling has been documented. Increased superficial tissue temperature results in the release of chemical mediators such as histamine and prostaglandins, which results in vasodilation and increased blood flow in some tissues. These vasodilatory mechanisms do not significantly affect blood flow in skeletal muscle since skeletal muscle blood flow is heavily influenced by other physiologic and metabolic factors. Exercise is the best means to increase blood flow to skeletal muscle.

Physiologic Effects

Many of the local physiologic effects of heat and cold have been studied thoroughly. Heat increases skin and joint temperature, improves blood circulation and muscle relaxation, and decreases joint stiffness. Cold numbs pain, decreases swelling, constricts blood vessels, and blocks nerve impulses to the joints (Oosterveld and Rasker, 1994; Sluka *et al.*, 1999; Brosseau *et al.*, 2003).

Deep heating, such as through thermal ultrasound (Dorn, 2015), is thought to lessen nerve sensitivity, increase blood flow, increase tissue metabolism, decrease muscle spindle sensitivity to stretch, cause muscle relaxation, and increase flexibility. Heat stimulates the cutaneous thermoreceptors that are connected to the cutaneous blood vessels, causing the release of bradykinin, which relaxes the smooth muscle walls resulting in vasodilation. Muscle relaxation occurs as a result of a decreased firing rate of the gamma efferents (sending signals away from the central nervous system), thus lowering the threshold of the muscle spindles and increasing afferent activity. There is also a decrease in firing of the alpha motorneuron to the extrafusal muscle fiber, resulting in muscle relaxation and decrease in muscle tone (Prentice, 1982; Peres *et al.*, 2002).

Veterinary Evidence for Both Hot and Cold Therapy

Evidence-based medicine for veterinary physical rehabilitation comments that heat is used to increase blood flow, increase collagen extensibility, and perhaps provide some mild analgesia (Millis and Ciuperca, 2015). A study of 10 healthy dogs that measured tissue temperatures measured at various depths of the lumbar, epaxial region found that application of a warm compress should be performed for 10 minutes (Millis and Ciuperca, 2015).

Cryotherapy is often applied in the early postinjury or postoperative period to reduce

blood flow, inflammation, swelling, and pain (Swenson *et al.*, 1996). Skin and superficial tissues are cooled to the greatest extent. Deeper tissues have a gradual decrease in tissue temperature during time exposed to cold. Rewarming after removal of the cold/ice pack is slower in deeper tissues. The rewarming rate of tissues depends on the duration of therapy and type of cryotherapy application. The greatest decrease in intra-articular temperature in dogs occurred with ice water immersion (Bocobo *et al.*, 1991). Rewarming of the canine stifle also took the longest after ice water immersion. The application of 10–20 minutes of ice water immersion caused a further significant temperature change at only the middle tissue depth; however, for maximal cooling, the minimum time of application should be 20 minutes. For postoperative extracapsular repair for cranial cruciate ligament rupture, cold compression and cold compression with bandaging were found to be equally beneficial in reducing stifle swelling in the first 72 hours. Cold compression was applied for 20 minutes by wrapping the leg from the stifle to the hock with a large cold pack and holding it in place with an elastic bandage once daily. Commercial canvas packs kept in a freezer or Ziploc® bags of ice wrapped in a towel are often used for 10–20 minutes every 6–8 hours while the patient is hospitalized (Corti, 2014).

General Recommendations for Heat

Heat should be applied for no more than 15–20 minutes at a time. Hot packs are relatively safe because they cool during treatment, minimizing the risk of burns. Padding is applied around the pack. The pack retains heat for approximately 30 minutes. The heat is absorbed mostly by the skin and subcutaneous fat. Wrapping of the hot pack with a towel or drape material is essential so the pack does not come in direct contact with the patient's skin. Heat wraps are marketed for people. Some products provide up to

Box 14.7 Contraindications for thermotherapy

- Electric heating pads and infrared lamps have a higher risk of burns. Electric heating pads should never be placed under an anesthetized patient or a patient with decreased superficial sensation. In general, a patient should never be left unattended during treatment, and the skin should be monitored frequently
- Do not use heat in an actively bleeding patient
- Acute inflammation
- Cardiac insufficiency
- Decreased impaired circulation in the area to be treated (to avoid overheating)
- Neurologic patients may not feel that it is becoming too hot
- Fever
- Malignancy
- Poor body heat regulation
- Pregnancy

Source: Batavia 2004. Reproduced with permission of Elsevier.

8 hours of continuous low level heat and could be used for small animals. Whirlpools have the advantage of also providing increased hydrostatic pressure to submerged body parts, which helps to increase lymphatic and venous flow from a distal-to-proximal orientation. Agitation within a whirlpool decreases the thermal gradient so that the temperature of the water in the tank is consistent throughout. The temperature of a whirlpool is based on the needs of the individual animal. For example, patients with chronic conditions may be treated with warmer water than patients with more acute disorders.

Precautions when Using Thermotherapy

Test the temperature of the modality you want to use or have the client test the temperature before applying to their pet. Heat packs especially can vary in temperature depending on ambient conditions. There is risk of overheating in dogs immersed in a heated whirlpool. They should be observed and their rectal temperature measured if in doubt. If there is reduced sensation because of nerve damage or even patient compromise in reaction time (sedation, reduced cognition), then caution should be used. Long-haired animals may heat less quickly than short-coat or clipped patients. A hot or cold pack can be heavy when placed on the animal,

so check for this by watching respirations or if the patient becomes irritable. In some instances, the pack can be placed under the area to be treated. Seek the advice of the supervising veterinarian if there are open or infected wounds (Box 14.7).

General Recommendations for Cold

A general rule for deciding when to apply cold versus heat is that cryotherapy should be used for the first 24–72 hours after acute injury when the acute signs of inflammation are present (swelling, redness, heat, and pain) (Steiss and Levine, 2005). If in doubt, use cold. If range of motion is decreased because of pain, apply cold; if range of motion is decreased because of stiffness, apply heat. Ice packs should be covered with a single layer of wet towel (moisture enhances heat exchange) or nothing between the skin and the ice pack. Otherwise, therapeutic temperatures may not be reached. Apply ice packs for up to 10–20 minutes. The duration may vary for different types of commercial ice packs. The recommendations on how often to apply ice vary. It is recommended not to apply cold to open wounds after 48–72 hours because of the vasoconstriction that occurs with cryotherapy. The supervising veterinarian should be asked for instructions concerning application of cold to individual patients.

Box 14.8 Contraindications for cryotherapy

- If there is a history of frostbite to the area, further cold application is contraindicated
- Cold should not be used in animals with generalized or localized vascular compromise or who possess an impaired thermoregulatory capacity

Precautions for Cold Therapy

The primary precaution is avoidance of frostbite. To be safe and to avoid prolonged application, use a timer and inspect the skin every few minutes. Observe for signs of frostbite during and after cryotherapy application (Millis, 2015).

Caution should be exercised when applying cryotherapy around superficial peripheral nerves because cases of cold-induced nerve palsy of the ulnar and superficial peroneal nerves have been reported in humans (Fox, 2014). Use caution in applying cold over open wounds, areas of poor sensation, or in very young or old dogs (Box 14.8).

A Word About Cats

Often it is assumed that cats will not tolerate rehabilitation due to temperament, lifestyle, or age. In fact, cats respond well to rehabilitation techniques, and often enjoy the mental stimulation of therapeutic exercise, stretching, and massage in the hospital and at home (Wright and Rychel, 2013).

Cryotherapy can be applied to regions of inflammation, particularly with an acute or chronic injury, to reduce hyperalgesia and slow local metabolism (Sluka *et al.* 1999). Heat therapy is also valuable, and often appreciated by feline patients. Application of a superficial hot pack over a tense muscle or region of chronic discomfort can help ease muscle spasm, improve blood flow, and reduce pain-associated behaviors (Sluka *et al.*, 1999; Lane and Latham, 2009).

Heat for Cats

Most cats like the application of heat, especially the use of infrared lamps, whereas the use of hot packs is sometimes difficult because cats often object to lying quietly with a hot pack on a joint or the back (Drum *et al.*, 2015). Many sociable cats prefer to lie in a lap or be held while hot packs are applied. A swaddling technique with a towel is useful as a low-stress restraint technique for hot-pack application.

Cold for Cats

It may be difficult to place the cold pack on the target area because cats sometimes refuse to lie quietly for more than a few minutes. Nevertheless, the therapist should try to use cold, especially in the early phase after surgery.

Conclusion

Selection of the appropriate modality depends largely on an understanding of the diagnosis, an accurate assessment of the stage of tissue healing and repair, an accurate clinical assessment of the functional limitations, the established treatment goals, and continued reevaluation of the patient. The physical rehabilitation therapist/practitioner will be the one deciding which therapy should be applied and when. The rehabilitation veterinary technician/nurse will be carrying out the instructions that the veterinarian provides. Cryotherapy is most useful during the acute inflammatory stages of tissue healing to cause vasoconstriction and to decrease edema and pain (Hanks *et al.*, 2015). Using heat too soon in the inflammatory process may exacerbate the inflammatory process and slow healing. Heating modalities (both superficial and deep) are most commonly used to cause vasodilation and increase tissue extensibility, and to decrease pain and muscle spasm (Hanks *et al.*, 2015).

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Modalities Part 2: Laser Therapy

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Introduction

Lasers were first used surgically in high doses to cut or seal tissue. The first report of lasers used for a biomodulatory (therapeutic) purpose was in 1963, by a team of researchers in Boston (Tunér *et al.*, 2007). This group used a laser to shrink tumors in hamsters, but the mechanism of this action was unknown. Shortly thereafter, Endre Mester, a Hungarian scientist, published several reports describing the ability of lasers to promote cutaneous wound healing in rats. It was also observed at this time that too little laser radiation would produce no effect and too much would be detrimental to healing. In 1967, Carney *et al.* showed that irradiation of skin tissue cultures resulted in increased collagen production. Thousands of others reports have since been

published, both supporting and refuting the effects of laser therapy. It is only recently that therapeutic lasers have gained wide recognition in veterinary medicine.

The purpose of this chapter is to describe basic laser physics, discuss the mechanisms behind “photobiomodulation” (modifying a biologic process by application of light), and describe the practical use of lasers in veterinary rehabilitation patients.

Laser Physics 101

Lasers, along with the sun, ordinary light bulbs, x-ray machines, and microwave ovens, emit electromagnetic radiation. Energy from these sources travels at the speed of light in packets known as photons. Photons travel in

waves, not unlike sound waves, and the type of radiation is distinguished by its wavelength. Electromagnetic radiation is typically described as a spectrum from very short (gamma rays, 1000–1 fm) to long (radio waves, 1000–1 m) wavelengths. In the middle of this spectrum lies visible light (400–800 nm) and infrared (1000–0.8 μ m), and laser radiation falls within these wavelengths (Tunér *et al.*, 2007).

Laser is an acronym for “light amplification by stimulated emission of radiation.” In this process of light amplification, electromagnetic energy is harnessed into an intense, coherent, monochromatic beam of light. The properties of monochromaticity (all waves are same length/color) and coherence (all waves in phase) are characteristics of all lasers. This is in contrast to light-emitting diodes (LEDs), which do not emit coherent light. The biomodulatory effects of lasers are believed to be due, in part, to their coherence, but there is some limited evidence to suggest that LEDs are as effective as lasers in photobiostimulation (Lubert *et al.*, 2000).

A laser is made using a material (gas, liquid, solid) that when stimulated by an external energy source, such as electricity, will release photons of a single color or wavelength. For example, a helium–neon (HeNe) laser will emit only wavelengths of approximately 633 nm, which fall within the visible light spectrum and produce the color red. Gallium arsenide (GaAs) lasers emit wavelengths of 904 nm, and are thus invisible within the infrared portion of the spectrum.

It is important to note that wavelength is inversely proportional to energy. This can be appreciated by recognizing differences between harmless radio waves (long wavelength, low energy) and gamma rays that are used in cancer radiation therapy (short wavelength, very high energy). Wavelengths between 300 and 400 nm represent UVA and UVB rays (Tunér *et al.*, 2007). These rays possess dangerous ionizing properties that cause sunburn. The wavelengths of therapeutic lasers typically all fall between 650 and 980 nm (Tunér *et al.*, 2007).

Therapeutic lasers are commonly referred to as “low-level lasers” or “cold lasers,” in contrast to high-powered lasers that are used to cut tissue. Therapeutic or low-level lasers have a power output less than 500 milliwatts (mW) and cannot cut tissue. Power output is significant, because a laser with a higher wattage will reach the desired dose more quickly. Among low-level lasers, power output (watts, W) varies greatly, anywhere from 3.5 mW to 500 mW. Lasers with power output greater than 500 mW (typically as high as 10 W) are available as therapeutic lasers. These lasers are not considered “low-level” or “cold lasers” because they have the capacity cause considerable tissue heating.

There are several other parameters that can differ between lasers. Lasers can be continuous or pulsed. When a laser is pulsed, the power output will reach a peak and return to zero at varying frequencies (Hz), or duty cycles. Therefore, the amount of power delivered will be the average power output.

Power density, or intensity, is the amount of power concentrated on a given area and is measured in W/cm². If the laser beam is spread over a larger area (larger spot size), the amount of energy at each point becomes less, compared to concentrating the energy at a single, small point (Mester *et al.*, 1968). This will be influenced by the beam diameter (mm) and spot size (cm²) that are specific to each laser.

Laser Classes

Lasers are divided into four safety classes with additional subclasses (I, II, IIIa, IIIb, and IV). A common misconception is that these classes distinguish the efficacy or quality of the laser. Rather, laser class is determined by the ability to cause eye injury and is based on power output, parallelism, diameter of beam, exposure time, and wavelength. Class I–IIIa lasers include supermarket scanners, laser pointers, and remote controls. Class IIIb lasers pose a risk of eye injury, and eye protection is recommended (Mester *et al.*, 1968).

Any laser with greater than 500 mW of average power falls into class IV, which is considered to be an acute hazard to the skin and eyes from direct and scattered radiation.

Dosage (also known as energy density or fluence) appears to be the most important laser parameter in clinical application. Dosage is measured in joules per centimeter squared (J/cm^2), and can be calculated using the following formula:

$$\text{Dose} = \frac{P \times t}{A}$$

where P is the laser's output power (W), t is treatment time (seconds), A is area treated (cm^2), and 1 joule = 1 W/s.

The usefulness of this formula is evident: By adjusting treatment time in relationship to a laser's output power, a practitioner can deliver the intended dosage. However, it should not be assumed that this relationship is linear; that is, faster delivery of an intended dose may not always equate with the same outcome as the equivalent amount of energy delivered over a longer period of time. This is an area of research that will have important implication in veterinary patients.

Lasers that are available commercially and marketed for medical or veterinary use will likely come with recommended doses for various conditions preprogrammed into the unit or listed in the instruction manual. Although the optimal treatment dose has not been established for any condition, generally recommended doses fall between 2–10 J/cm^2 of energy recommended at the target tissue depth.

One last consideration is the depth of laser penetration. This depends on the wavelength of the laser, with longer wavelength resulting in deeper penetration. A GaAs laser (904 nm) can reach tissue depths of 3–5 cm, while a HeNe (633 nm) will have more superficial penetration, near 1 cm (Corazza *et al.*, 2007). Other factors may influence the depth of penetration, including hair coat, skin pigmentation, and tissue composition (Bjordan *et al.*, 2006b). Much of the radiation

energy may be absorbed by hair, dark pigmented skin, and highly vascular tissue such as muscle, so energy is attenuated before it reaches deeper tissue. Energy output from class IV lasers with heating capacity can also be absorbed by fat, which acts as a thermal insulator. This has to be taken into consideration by your supervising veterinarian when treatment dose is calculated. Note, unlike sound waves, laser waves (electromagnetic radiation) can penetrate bone.

Photobiomodulation: Proposed Mechanisms of Action

Photobiomodulation is the application of light in order to modify a biologic process. Depending on the wavelength, power, and other factors, light can cause beneficial or harmful effects in cells and tissue (Bjordan *et al.*, 2006a). Therapeutic lasers can be used to stimulate favorable effects in tissue, including enhanced wound healing and modulation of pain (Medrado *et al.*, 2008). There have been numerous experimental animal and human studies investigating the mechanisms of action of lasers in these conditions (Hawkins and Abrahamse, 2006). Currently, there are very few peer-reviewed published research studies of laser use in small animal veterinary patients, so we are left to extrapolate how lasers work and how best to use laser therapy from the currently available evidence.

Laser therapy can affect each stage of tissue healing (inflammation, proliferation, maturation) (da Silva *et al.*, 2010). The role of laser in the inflammatory phase is that of immunomodulation. Some studies that have shown enhancement of the inflammatory response and increased production of growth factors such as transforming growth factor beta 1 (TGF- β 1), while others show a reduction of inflammatory cytokines such as tumor necrosis factor alpha (TNF α) and prostaglandin E2 (PGE2) (Callies *et al.*, 2011). Studies have

demonstrated acceleration (shortening) of the inflammatory phase, with rapid progression into the proliferative phase of healing (Mester *et al.*, 1971). Furthermore, laser therapy has been shown to reduce inflammatory mediators, such as COX-2 and PGE, and to decrease inflammatory cell numbers in a tissue, and edema (Mester *et al.*, 1970). It is clear from a large number of studies that the proliferative phase of wound healing is greatly enhanced by therapeutic lasers (Mester *et al.*, 1968). Research conducted on cell cultures has found that various wavelengths and doses are effective at increasing fibroblast proliferation (Wong-Riley *et al.*, 2005). Similar findings have been demonstrated in research animal models of wound healing (Fulop *et al.*, 2009). However, Kurach *et al.* (2015) more recently reported no apparent benefits of low-level laser therapy in a model of acute wound healing in dogs. Furthermore, it has been shown that laser therapy can stimulate epithelialization and collagen deposition in animal models of diabetes and glucocorticoid excess, where wound healing is usually retarded (Woodruff *et al.*, 2004). The final phase of wound healing is maturation or remodeling of scar tissue. Lasers influence this phase by enhancing the organization of collagen fibers within wounds (Montesinos, 1988). While these results are promising, at the time of writing this chapter none of these effects have been confirmed in clinical veterinary patients.

The mechanisms by which laser decreases pain are not as well understood. One of the primary means of decreasing pain is by decreasing the production of inflammatory products. Laboratory studies have shown the ability of laser to decrease the production of inflammatory products such as PGE2 and TNF α , and inhibit COX-2 (Viegas *et al.*, 2007). Besides decreasing inflammation, laser therapy is believed to reduce pain through several other mechanisms that are particularly important when treating chronic pain. Photochemical reactions take place on a subcellular level and lead to increased ATP and stabilization of cell membranes and ATP-mediated pumps. This in turn results in

restoration of the cell membrane electrochemical gradient and resting membrane potential of nerve cells (Wakabayashi *et al.*, 1993). Excessive or inappropriate depolarization of peripheral nociceptors is minimized and the conduction of pain along C fibers is slowed (Sakurai *et al.*, 2000). Other postulated mechanisms of pain relief include increased release of beta-endorphins and serotonin, and enhanced removal of inflammatory mediators from the site injury due laser-induced changes in local hemodynamics (Mendez *et al.*, 2004).

It is important to note that a therapeutic window exists for photobiostimulation, which likely explains the discrepancy in results between various studies. At subtherapeutic doses, cells will not be stimulated and no reactions will occur; at extremely high doses, detrimental effects can be seen (Arany *et al.*, 2007). The modulatory effects may also be wavelength-specific and vary with other laser parameters such as polarization or pulse frequency.

Contraindications

Because laser is recognized to enhance neovascularization, irradiation of tumors or wounds that may contain cancer cells is considered contraindicated, despite there being a small body of evidence that laser can reduce tumor growth (Tunér *et al.*, 2007). Furthermore, lasers pose a known risk to the eye, so irradiation of or near the eye should not be performed. Additional contraindications include irradiation over a pregnant uterus and open growth plates (Oliveira *et al.*, 2012).

Laser Safety

While administering a laser treatment it is important to take precautions to protect yourself and the patient, as well as being mindful of others in the room. All class IIIB and IV lasers should come with their own set of safety glasses.



Figure 15.1 Wavelength-specific safety goggles. Note the stamp in the upper corner indicating wavelengths the goggles have been tested and approved for.

These glasses have wavelength-treated lenses that are specific to the laser. They will have stamping on the corner of the lens indicating what wavelengths are covered (Figure 15.1). It is important to note that no other shaded glasses can block out laser irradiation, including sunglasses or surgical laser glasses.

Some laser companies provide pet safety glasses as well, or “doggles.” Make sure they have the same wavelength stamping the human glasses do before relying on them to block laser during emission. As safe as they are stylish, it can be a challenge getting your patient to wear these without a struggle. If they refuse to wear the doggles, ensure their head is facing away from the laser beam. A towel or e-collar may be useful to keep patients from turning their head (Figure 15.2).

The ideal location for a laser treatment is a small, windowless exam room with a comfortable bed on the ground for you and your



Figure 15.2 This patient would not tolerate safety goggles so a soft, blue e-collar was used. Ideally the collar should be flipped up to shield the face completely from potential reflection or scatter from laser beam. Liz Brown in image.

patient to sit on. There should be a “Do Not Disturb; Laser in Progress” sign on the door. Everyone present in the room should be wearing safety glasses. If this scenario is impossible in your hospital make sure the laser operator and everyone within range of the beam is wearing their protective glasses. Then, take a look at your surroundings. Are there metallic surfaces nearby (e.g., a “wet” table)? If so, cover them with a towel or blanket (Figure 15.3).

Laser has the ability to reflect once it comes in contact with metal. This can be hazardous to others in the room that are not protected. For this reason, treating a patient inside of a stainless steel cage is strongly discouraged, as the risk for scatter is too great. Also, it is recommended to remove metallic rings, watches, bracelets, and dog collars or harness with tags and metallic claps or adornments (Figure 15.4).



Figure 15.3 This patient is being treated on a stainless steel table. Blankets and towels have been placed to cover the surface to minimize the risk of laser beam reflection from the steel surface. Ideally this patient should have on safety goggles or other method of protecting the eyes. Liz Brown in image.

Laser therapy is often performed over metallic orthopedic implants. In order to reduce reflection of the laser off the metallic implants within the body, the laser beam should be angled away from metal implants as much as possible. A laser that does not produce heat should be considered. It is essential that the technician or veterinarian administering laser therapy has a good understanding of anatomy and the location of orthopedic implants. Your supervising veterinarian will guide you.

There are not many federal regulations regarding therapeutic lasers in the United States. Those regulations that do exist for lasers are directed toward industrial lasers. At the state level, very few deviate from the federal regulations that are in place for industrial lasers. Your veterinarian should have a clinic policy regarding laser use and will need your help keeping that imple-



Figure 15.4 What *not* to do: This patient's collar has metallic adornments and tags. The laser is not in full contact with the skin and is angled toward the collar, resulting in invisible but hazardous reflection of radiation. When treating the neck and shoulder you should remove all metallic dog collars and ensure that the laser is held perpendicular to the tissue. Not all lasers can be held in contact with the skin (i.e., those that generate heat should not be used in contact mode).

mented. The Occupational Safety and Health Administration (OSHA) in the United States recommends a laser use log be kept, separate to the patient record. This will be similar to an x-ray log (https://www.osha.gov/dts/osta/otm/otm_iii/otm_iii_6.html).

Read the owner's manual carefully. Many companies will provide their own resources for safe operation.

- You can find more information on the consensus standard (Smalley, 2011) by getting a copy of ANSI Z136.1 from the Laser Safety Institute of America's website (<https://www.lia.org/store/ANSI%20Z136%20Standards>).
- For international standards, go to World Association of Laser Therapy (WALT; <http://waltza.co.za/>).

Using a Therapeutic Laser in Practice

There are many different lasers available in the veterinary market today, and while the basic mechanism of action is similar, they all function differently. User competency is a big variable in the outcome of your treatment, so it is important that each operator has been properly trained on the device before administering laser. One of the veterinarians in your practice should be assigned as the laser expert. This person will ensure that all new staff are trained and comfortable with the laser. An excellent knowledge of anatomy is crucial; the veterinarian needs to be able to get a realistic estimate of target tissue depth as well as likely attenuation levels. There are many supportive resources available online, including supportive websites from several veterinary laser manufacturers.

Once all staff have been trained, the next step is implementing the laser into daily practice. Treatments should be first prescribed by a veterinarian. Laser protocol and number of treatments are decided upon by the overseeing veterinarian, and then generally performed by a team member. Assistants can administer treatments as long as they have been trained on the machine and adhere to all safety precautions.

Positioning your patient can be tricky, since you will often be chasing a moving target. The more comfortable and relaxed the animal is, the easier your job will be. Most animals feel more secure on the ground instead of up on a table. Whether you are using the scanning or point-to-point technique, you will want to make sure the probe is kept at a 90-degree angle to your target area to ensure even and optimal laser emission and penetration (Figure 15.5).

Having knowledge of muscle anatomy and boney landmarks enables accurate treatment and makes it easier to follow your supervising veterinarian's instructions. A rehabilitation certification course will help you to understand bone and soft



Figure 15.5 The laser probe should always be perpendicular to the target tissue. Lasers that do not generate substantial heat can be placed in contact with the skin/hair and a point-to-point method used to administer treatment. Heat-generating/high-powered lasers should not be used in contact and may need to be continuously moved (scanning) to avoid burning the tissue. Even with scanning, the probe should remain at a 90-degree angle to the body surface.

tissue anatomy as it applies to laser therapy. For example, a 10-year-old Labrador who had a tibial plateau-leveling osteotomy (TPLO) surgery performed when he was 2 years old has now developed moderate osteoarthritis in the joint. The veterinarian has prescribed laser to help with the pain and inflammation and should have given you specific instructions for probe placement. But what if they did not? You have your diagnosis and your treatment plan of action. Now, where to place the probe? First you should be comfortable palpating and identifying the metal plate used in the surgery. Next you need to find the joint margins where the joint capsule attaches. The joint capsule is your primary target when treating osteoarthritis because it is the tissue that contains nerve endings and inflammatory mediators (Bjordal *et al.*, 2003). If you were treating a dog within the first few months after TPLO, your target tissues would be the joint capsule, as well as surgical incision, and the bone where the TPLO cut was made, remembering to direct your laser around the metal plate.

Table 15.1 Common conditions treated and protocols used by the authors. At the time of writing, there are no research studies confirming the optimal laser treatment protocols in veterinary patients; the protocols listed are based on WALT guidelines and the authors' clinical experience. It is important to note that these are guidelines and each patient is treated as an individual, including adaptations of protocols, inclusion of other treatment sites, and in all cases, additional rehabilitation modalities including manual therapy and therapeutic exercise.

Indication	Target tissue	Treatment frequency	Dose (J/cm ²)
Cranial cruciate ligament rupture: post surgery	Incision and areas of surgical tissue manipulation, joint capsule (medial and lateral), patellar tendon; sartorius/quadriceps if tight or atrophied	Immediately postoperatively and again the following day; then beginning 2 weeks post surgery, 2–3 times/week for a total of 8–10 treatments, then once monthly for maintenance. Repeat loading dose as needed	2–4
CCLR chronic or non-surgical	Joint capsule, sartorius, quadriceps, gastrocnemius origin	2–3 times/week for a total of 8–10 treatments, then once monthly for maintenance. Repeat loading dose as needed	2–8
Intervertebral disc disease (hemi-laminectomy; ventral slot)	Surgical incision, spinal column over lesion; articular facets surrounding lesion (plus 2 cranial and 2 caudal); surrounding thoracic, lumbar, and cervical muscles according to localization; consider treating hypertonic muscles, path sciatic and femoral nerves	Acute: 24 hours postoperatively and daily as long as hospitalized. Chronic: 2–3 times/week for 4–6 weeks or as long as needed based on recovery and rehabilitation plan	Acute: 2–4 Chronic: 4–6
Hip dysplasia	Gluteal muscles, pectineus, epaxials, quadriceps and hamstrings if tight or atrophied; joint capsule (may be too deep to reach in large dogs); ± LS space and SI	2–3 times/week for a total of 8–10 treatments, then once monthly for maintenance. Repeat loading dose as needed	4–10
Shoulder tendinopathies	Supraspinatous insertion and musculotendinous junction, origin of biceps tendon ± belly of biceps and insertions at elbow; deltoids, triceps muscles if tight or atrophied	2–3 times/week for a total of 8–10 treatments, then once monthly for maintenance. Repeat loading dose as needed	4–8

Table 15.1 lists common uses of therapeutic laser in veterinary rehabilitation, the target tissue(s), and protocols used by the authors.

Conclusion

Integrating laser into physical rehabilitation can be tremendously rewarding. Often times the patient will even enjoy the session. Like other treatment, it will work wonders for some patients while others may not display obvious signs of benefit. At

this time, there is very little research in clinical veterinary patients as to the optimal laser treatment protocols. The World Association of Laser Therapy (<http://waltza.co.za/>) provides recommended treatment protocols for class IIb lasers for treating a variety of musculoskeletal conditions in human patients. Many veterinary therapeutic lasers are preprogrammed with treatment settings, and some are based on the WALT recommendations. Until more research is available that identifies the best treatment protocols in veterinary patients,

we can use the WALT guidelines, laser company settings, and ultimately a “do no harm” approach to guide our laser treatments. Frequent veterinary reassessment is need to monitor the effectiveness of laser therapy and to modify treatment as

needed. The veterinary technician/nurse can help assessment of therapeutic effect by taking an accurate history of home behavior since last therapy visit and by carefully palpating the affected area to monitor pain level (analgesic effect).

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Modalities Part 3: Electrotherapy and Electromagnetic Therapy

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Introduction

Electrotherapies are commonly utilized in canine rehabilitation. Electrotherapy uses electrical stimulation to apply therapeutic levels of electric currents in various and specific waveforms to decrease muscle atrophy, assist in muscle re-education, improve muscle tone, reduce edema, facilitate wound healing, and reduce pain. When used appropriately, this non-pharmacological therapy can be a valuable and effective adjunct to the treatment plan.

The terminology associated with electric stimulation can be confusing. It is a good idea to review specific definitions pertaining to electrotherapy.

Electrical Stimulation

Electrical stimulation can be used as an adjunct to manage many disorders, including musculoskeletal, soft tissue, vascular, and neuromuscular injuries (Hecox *et al.*, 1994). The most commonly used forms of electrical stimulation are neuromuscular electrical stimulation (NMES) – a term used interchangeably with electrical stimulation, transcutaneous electrical nerve stimulation (TENS), and functional electrical stimulation (FES). Alternating current (AC), is an electric current in which the flow of electric charge periodically reverses direction, whereas in direct current (DC, also dc), the flow of electric charge

is only in one direction. There are three types of therapeutic electrical stimulation currents: direct (or monophasic) current, alternating (or biphasic) current, and pulsed current (may be mono- or biphasic) (Niebaum, 2013).

To use terminology accurately, almost all electrical stimulators are TENS units, as they work transcutaneously via surface electrodes to excite nerves. In the typical scenario where the muscle is innervated by a motor nerve, NMES is the appropriate terminology (as it is the nerve supplying that muscle that is stimulated in order to activate the muscle), and when a muscle is denervated and requires direct muscle fiber activation through electrical stimulation, the term EMS is used (Levine and Bockstahler, 2014). Functional electrical stimulation (FES) is a technique that uses electrical currents to activate nerve trunks innervating extremities (more than one muscle innervated) affected by paralysis resulting from spinal cord injury, head injury, stroke, and other neurological disorders (Delitto and Robinson, 1989). FES is primarily used to restore function in people with disabilities. It is sometimes also referred to as neuromuscular electrical stimulation (NMES) (Ferrari de Castro and Cliquet, 2000).

Direct Current

Direct current (DC) electrical stimulation is used to stimulate denervated muscles, for wound healing, and for iontophoresis. It creates a unidirectional flow of charged particles with a waveform usually using 20–200 microamperes at a low voltage (Gardner *et al.*, 1999; Cameron, 2003).

Alternating Current

Alternating current (AC) is a bidirectional flow of charged particles using low-voltage milliamperage (mA). The charges are equal in the two symmetrical phases of each pulse (a pulse is a burst of electromagnetic energy). Therefore, the accumulation of charge in the tissues is zero. TENS is an example of an AC

Table 16.1 Recommended pulse duration for neuromuscular electrical stimulation in dogs.

Muscle	Recommended pulse duration in dogs (ms)
Supraspinatus	150–190
Infraspinatus	130–170
Deltoidaeus	160–200
Triceps brachii	200–240
Gluteus medius	160–200
Biceps femoris	180–220
Semitendinosus	150–190
Vastus lateralis	210–250
Tibialis cranialis	210–250
Erector spinae	180–220

Source: Adapted from Sawaya *et al.* (2008).

therapy. Alternating currents are delivered at various pulse widths depending on the muscle. Recommendations have been made for dogs based on research (Johnson *et al.*, 1997; Sawaya *et al.*, 2008) (Table 16.1).

Pulsed Current

The output of this therapy is one of short paired pulses with a long interval of no pulses between and is typically delivered at 75–200 V and 80–100pps. The current flow may be one direction (like DC) or bidirectional (like AC). If one direction, it is referred to as monophasic pulsed current. If current flows in two opposite directions, it is referred to as biphasic pulsed (Gardner *et al.*, 1999; Cameron, 2003).

Basic Terminology

To help the technician/nurse understand the benefits that electrical stimulation may provide in veterinary practice, a basic explanation and definition of the electrical current parameters (Canapp, 2007) used are summarized in Box 16.1. Each session lasts between 5 and 20 minutes depending on the degree of muscle fatigue (Prydie and Hewitt, 2015).

Box 16.1 Current classification

There are three basic waveforms used in commercial therapeutic electrical stimulation units: direct current, alternating current, and pulsed current.

Direct current (DC)

- Continuous unidirectional flow of charged particles with a duration of at least 1 s
- One electrode is always the anode (+) and one is always the cathode (–) for the entire event
- There is a build-up of charge since it is moving in one direction causing a chemical effect on the tissue under the electrode

Alternating current (AC)

- Uninterrupted bidirectional flow of charged particles changing direction at least once per second
- Electrodes continuously changes polarity each cycle, therefore no build-up of charge under the electrodes
- Often used in interferential or Russian commercial stimulators

Pulsed current (pulsed)

- Can be unidirectional (like DC) or bidirectional (like AC)
- Flow of charged particles stops periodically for less than 1 s before the next event
- Pulses can occur individually or in a series

Amplitude or Intensity

Measured in milliamperes (mA), amplitude is the amount of the current being delivered and describes the total magnitude of the electrical wave. A higher amplitude will result in a stronger muscle contraction due to recruitment of a greater number of muscle fibers. Higher intensities have been shown to promote strength gains in people (Gondin *et al.*, 2011).

Pulse Duration or Pulse Width

Pulse width is measured in microseconds (μ s). Electricity produced in electrotherapy flows in a waveform, each wave corresponding to a rise in the current of electron flow followed by an abrupt cessation, or a slow decline in electron flow to zero before the next wave follows. The time during which the electric current flows is known as the pulse duration or width. A longer pulse duration may stimulate more nerve fibers that mediate pain, but has the benefit of requiring a lower amplitude of current to achieve a muscle contraction (Niebaum, 2013). Shorter pulse widths tend to be more comfortable,

but a longer pulse width can potentially recruit more muscle fibers to compensate for fatigue (Lagerquist and Collins, 2010). Wider pulse widths (300–400 ms) have been shown to produce stronger muscle contractions (Badylak *et al.*, 1990) and penetrate to deeper layers of muscle tissue. So, when trying to affect deeper tissue layers, a longer pulse width should be used (Bracciano, 2008). However specific pulse durations have been calculated for comfort and effectiveness in the dog (Sawaya *et al.*, 2008) (see Table 16.1).

Frequency or Pulse Rate

Frequency is measured in hertz (Hz) and is the number of pulses produced per second (pps) during the time the current is flowing. The best frequency will vary depending on the goal of therapy. Frequencies as low as 20 Hz will produce tetanic (whole sustained) muscle contractions (Spurgeon *et al.*, 1978; Levine and Bockstahler, 2014) but only sub-maximal forces of contraction are typically produced in this range. Maximal contraction force is generally achieved at frequencies between 60 and 100 Hz in humans (Duchateau and Baudry, 2014). As frequency